Plant Biotechnology Journal

Plant Biotechnology Journal (2024) **22**, pp. 617–634 doi: 10.1111/pbi.14210

The chromatin remodeller MdRAD5B enhances drought tolerance by coupling MdLHP1-mediated H3K27me3 in apple

Yi Song[1,†], Jieqiang He[1,†], Junxing Guo[1,†], Yinpeng Xie[1], Ziqing Ma[1], Zeyuan Liu[1], Chundong Niu[1], Xuewei Li[1], Baohua Chu[1], Muhammad Mobeen Tahir[1], Jidi Xu[1], Fengwang Ma[1] and Qingmei Guan[1,2,*]

[1]*State Key Laboratory of Crop Stress Biology for Arid Areas/Shaanxi Key Laboratory of Apple, College of Horticulture, Northwest A&F University, Yangling, China*
[2]*Shenzhen Research Institute, Northwest A&F University, Shenzhen, China*

Received 12 June 2023;
revised 25 September 2023;
accepted 6 October 2023.
**Correspondence* (Tel +86 13700296910; fax +86 029 87082864; email qguan@nwsuaf.edu.cn)
[†]These authors contributed equally to this work.

Keywords: MdRAD5B, chromatin accessibility, MdLHP1, ubiquitin-independent 20S proteasome pathway, H3K27me3.

Summary

RAD5B belongs to the Rad5/16-like group of the SNF2 family, which often functions in chromatin remodelling. However, whether RAD5B is involved in chromatin remodelling, histone modification, and drought stress tolerance is largely unclear. We identified a drought-inducible chromatin remodeler, MdRAD5B, which positively regulates apple drought tolerance. Transposase-accessible chromatin with high-throughput sequencing (ATAC-seq) analysis showed that MdRAD5B affects the expression of 466 drought-responsive genes through its chromatin remodelling function in response to drought stress. In addition, MdRAD5B interacts with and degrades MdLHP1, a crucial regulator of histone H3 trimethylation at K27 (H3K27me3), through the ubiquitin-independent 20S proteasome. Chromatin immunoprecipitation-sequencing (ChIP-seq) analysis revealed that MdRAD5B modulates the H3K27me3 deposition of 615 genes in response to drought stress. Genetic interaction analysis showed that MdRAD5B mediates the H3K27me3 deposition of drought-responsive genes through MdLHP1, which causes their expression changes under drought stress. Our results unravelled a dual function of MdRAD5B in gene expression modulation in apple in response to drought, that is, *via* the regulation of chromatin remodelling and H3K27me3.

Introduction

Drought is a critical factor for plant distribution, growth, and productivity and causes huge economic loss in agriculture (Chen *et al*., 2022; Geng *et al*., 2018; Gupta *et al*., 2020; Talbi *et al*., 2015). To cope with drought stress, plants have developed several strategies at morphological, physiological, and molecular levels (Gupta *et al*., 2020). Plants can accumulate more sugars, amino acids, proline, enzymatic, and non-enzymatic antioxidant enzymes in response to drought stress at the physiological level (Abid *et al*., 2016; Kang *et al*., 2002; Niu *et al*., 2013; Saneoka *et al*., 2004). At the molecular level, abundant drought-responsive genes characterized as transcriptional factors, structure genes, and epigenetic factors are involved in modulating drought tolerance, revealing the potential perspective of drought-tolerant crop improvement (Dong *et al*., 2014; Kang *et al*., 2002; Ramegowda *et al*., 2017). In general, plants respond to drought by altering gene expression through multiple mechanisms, including chromatin remodelling, post-translational modifications on histones or proteins, modification of DNA itself, and the architecture of gene promoters driven by different *cis-elements* (Dong *et al*., 2014; Kumar *et al*., 2016; Ramegowda *et al*., 2017).

SNF2 (Sucrose non-fermenting 2) family proteins can use energy from ATP hydrolysis to facilitate DNA translocation. They can therefore alter the accessibility of DNA by remodelling chromatin structure (Clapier and Cairns, 2009; Ryan and Owen-Hughes, 2011). Depending on the chromatin remodelling function, SNF2 proteins always play critical roles in plant development and response to different hormones and abiotic stresses (Hu *et al*., 2013; Knizewski *et al*., 2008; Sarnowska *et al*., 2016). Arabidopsis RAD5B, belonging to the Rad5/16-like group, have similar HIRAN/SNF2_N/RING/HELICASE_C domain architectures to yeast RAD5. Arabidopsis RAD5B is involved in reproductive and root development (Khatri *et al*., 2017). The homologue of RAD5B in cucumber SH1 can regulate LDUVB-dependent hypocotyl elongation by modulating the UVR8 signalling pathway (Bo *et al*., 2016). Whether RAD5B is involved in chromatin remodelling and response to different stresses is largely unknown.

Histone modifications, referring to the posttranslational covalent modifications on the tails of core histones, play an important role in gene transcriptional regulation by modulating chromatin status (Duan *et al*., 2018; Yuan *et al*., 2013). Among different histone modifications, histone methylation has been widely studied due to its critical role in various biological processes, including transcriptional regulation of genes and TEs involved in plant development and stress response (Liu *et al*., 2010). Plant histone methylation occurs in histone H3 at four lysine sites, including 4, 9, 27, and 36. Generally, H3K4 and H3K36 methylation are considered gene activation marks, and H3K9 and H3K27 methylation are associated with gene repression (Liu *et al*., 2010). The histone methylation process is controlled by a series of “writers” (methyltransferases), “erasers” (demethylases), and “readers”. In particular, H3 lysine 27 trimethylation (H3K27me3) is achieved by two polycomb group (PcG) repressive complexes (PRCs), PRC1 and PRC2, which were first described in Drosophila (Gan *et al*., 2013, 2015). LIKE HETEROCHROMATIN PROTEIN 1/TERMINAL FLOWER 2 (LHP1/TFL2), considered a component of the PRC1 complex, can bind H3K27me3 *in vivo* and *in vitro* and show co-localization with H3K27me3 at the genome-wide level (Ramirez-Prado *et al*., 2019; Turck *et al*., 2007; Zhang *et al*., 2007). LHP1 achieves its regulatory functions by binding to H3K27me3 marks and repressing the PRCs targets, such as FLOWERING LOCUS C (FLC), FLOWERING TIME (FT), MYC2, and AGAMOUS (AG) (Kotake *et al*., 2003; Libault *et al*., 2005; Ramirez-Prado *et al*., 2019; Turck *et al*., 2007). Additionally, LHP1 can interact with the subunit of PRC2, MULTICOPY SUPPRESSOR OF IRA1 (MSI1), thereby recruiting PRC2 to repress gene expression (Derkacheva *et al*., 2013).

Apple is one of the most important economic fruit crops cultivated in temperate regions around the world. Under the background of climate change, apple trees always suffer from drought stress because of the water deficit in the growing season. To date, molecular breeding has become a useful method for increasing apple drought tolerance (Chen *et al*., 2022; Geng *et al*., 2018; Li *et al*., 2020a). Therefore, it is critical to understand the underlying mechanisms of apples in response to drought stress. Although physiological changes, transcriptional regulations, and epigenetic regulations in response to drought have been investigated in apples (Chen *et al*., 2020, 2022; Geng *et al*., 2018; Hou *et al*., 2022; Jiang *et al*., 2022; Li *et al*., 2020a, 2022a, b; Liu *et al*., 2021; Niu *et al*., 2019; Shen *et al*., 2022; Xie *et al*., 2018, 2021; Xu *et al*., 2018), the involvement of chromatin remodelling in response to drought remains elusive. Here, we report a chromatin remodelling factor, MdRAD5B, which enhances the drought tolerance in apple by modulating the chromatin accessibility and H3K27 trimethylation deposition on drought-responsive genes. These findings gain a new insight into the mechanism of apple responding to drought stress and provide candidate genes for drought-tolerant crop improvement.

Results

MdRAD5B plays a positive role in drought tolerance

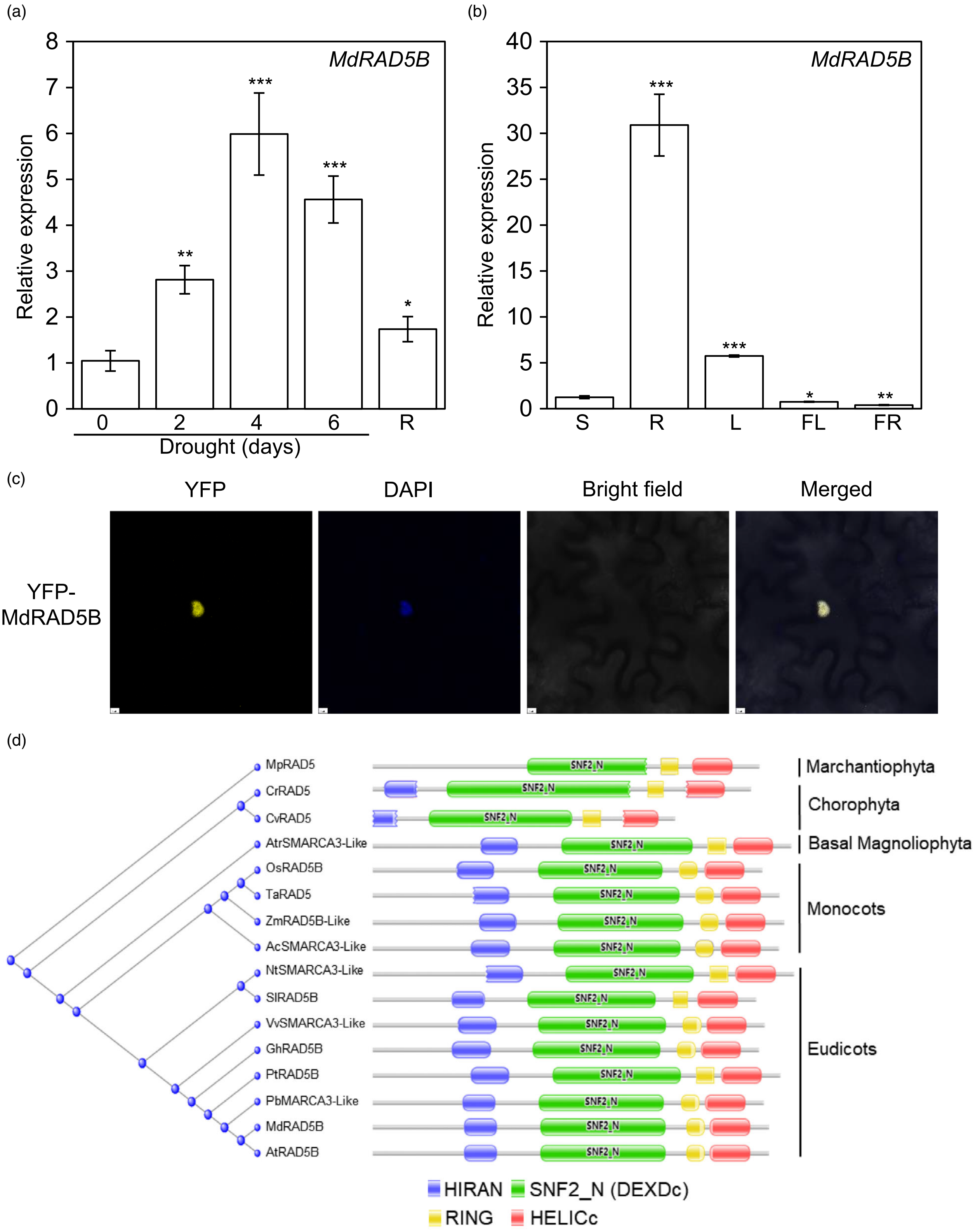
To understand if the homologous gene of *RAD5B* plays a role in drought response in apple, we examined its expression in *Malus domestica* cv. Golden Delicious treated with drought stress. Results showed that *MdRAD5B* could be induced significantly by drought treatment (Figure 1a), indicating that *MdRAD5B* might play a role in apple drought response. Organ-specific expression analysis revealed that *MdRAD5B* mRNA was abundant in the roots and leaves (Figure 1b), consistent with its potential role in drought response. Subcellular localization analysis showed that MdRAD5B was localized in the nucleus (Figure 1c). We then cloned *MdRAD5B* from *M. domestica* cv. Golden Delicious. Multiple sequence alignment and domain analysis revealed that MdRAD5B contains a similar architecture to RAD5B (Figure S1). Phylogeny analysis showed that RAD5B proteins are conserved across plant species from Chlorophyta to Magnoliophyta (Figure 1d).

To examine the biological function of MdRAD5B in drought response, we generated the *MdRAD5B* overexpression (OE) transgenic lines and RNA interference (RNAi) lines (Figure S2). After being transferred to soil for 60 days, the non-transgenic (GL-3) and *MdRAD5B* RNAi plants were treated with short-term drought stress. We found that 24% of GL-3 plants survived, whereas only 10% of *MdRAD5B* RNAi plants were still alive after drought stress, followed by recovery with rewatering (Figure 2a, b). Drought stress increased electrolyte leakage and the malondialdehyde (MDA) content in all the plants. However, *MdRAD5B* RNAi transgenic lines had significantly higher ion leakage and MDA content than GL-3 plants under drought stress (Figure 2c,d), indicating more severe membrane damage of *MdRAD5B* RNAi plants as compared with GL-3 plants. In addition, *MdRAD5B* RNAi plants lost water more quickly than GL-3 plants and had higher stomatal conductance under drought stress (Figure S3a,b). The GL-3 and *MdRAD5B* OE plants were also treated with short-term drought stress after being transferred to soil for 45 days. On the contrary, *MdRAD5B* OE transgenic plants had a significantly higher survival rate than GL-3 plants after drought stress (Figure 2e,f). Moreover, ion leakage and MDA content were lower in *MdRAD5B* OE transgenic plants than in GL-3 plants under drought stress (Figure 2g,h). In addition, *MdRAD5B* OE plants had lower water loss rate and stomatal conductance than GL-3 plants (Figure S3c,d). These results suggest that MdRAD5B is a positive regulator for apple drought tolerance.

MdRAD5B regulates the expression of drought-responsive genes

SNF2 proteins are necessary for the inducible or repressible transcription of certain genes (Clapier *et al*., 2017). We were next interested in whether MdRAD5B could globally regulate gene expression under control and drought stress. Previous study showed that cellular water loss marks the first event of drought stress (Gupta *et al*., 2020), we therefore used dehydration treatment to simulate drought stress and then performed RNA-seq of *MdRAD5B* RNAi and GL-3 plants under dehydration conditions. RNA-seq analysis of *MdRAD5B* RNAi and GL-3 plants revealed a total of 406 and 5209 genes ($|\log_2 FC| > 1$, *P*-adjust <0.01) were differentially expressed in *MdRAD5B* RNAi plants under control and dehydration conditions, respectively, as compared with GL-3 plants (Data S1–S3). Among the 406

Figure 1 MdRAD5B is responsive to drought stress and localized in the nucleus. (a) Time-course expression of *MdRAD5B* in *Malus domestica* in response to drought stress. R, rewater. (b) Organ-specific expression pattern of *MdRAD5B* in *Malus prunifolia*. R, root; S, stem; L, leaf; FL, flower; FR, fruit. (c) MdRAD5B is localized in the nucleus of tobacco leaf epidermal cells. Bars = 5 μm. (d) Phylogenetic tree of RAD5B protein from different species. *Marchantia polymorpha*, *MpRAD5*, PTQ49827.1; *Chlamydomonas reinhardtii*, *CrRAD5*, PNW72987.1; *Chlorella variabilis*, *CvRAD5*, XP_005846452.1; *Amborella trichopoda*, *AtrSMARCA3-like*, XP_006851757.3; *Oryza sativa*, *OsRAD5B*, XP_015634613.1; *Triticum aestivum*, *TaRAD5*, KAF7006320.1; *Zea mays*, ZmRAD5B-Like, NP_001333874.1; *Ananas comosus*, *AcSMARCA3-like*, XP_020084072.1; *Nicotiana tabacum*, *NtSMARCA3-like*, XP_016462752.1; *Solanum lycopersicum*, *SlRAD5B*, XP_004234234.2; *Vitis vinifera*, *VvSMARCA3-like*, XP_002280814.1; *Gossypium hirsutum*, *GhRAD5B*, XP_016715833.2; *Populus trichocarpa*, *PtRAD5B*, XP_006385564.1; *Pyrus bretschneideri*, *PbSMARCA3-like*, XP_018501035.1; *Malus domestica*, *MdRAD5B*, XP_008341115.2; *Arabidopsis thaliana*, *AtRAD5B*, NP_199166.1. SMARCA3, putative SWI/SNF-related matrix-associated Actin-dependent regulator of chromatin subfamily A member 3. Error bars indicate standard deviation [n = 3 in (a, b)]. Student's t test was performed and statistically significant differences were indicated by *$P < 0.05$, **$P < 0.01$, or ***$P < 0.001$.



differentially expressed genes (DEGs) in *MdRAD5B* RNAi plants, 344 were up-regulated while 62 were down-regulated, as compared with GL-3 plants (Data S2). Among the 5209 DEGs, 3694 DEGs were up-regulated while 1515 DEGs were down-regulated, as compared with GL-3 plants under dehydration conditions (Figure 3a,b; Data S3). These data imply that MdRAD5B can positively or negatively regulate the expression of genes at the whole transcriptomic level in response to drought stress. Moreover, we found that 1997 drought-repressed genes were up-regulated in *MdRAD5B* RNAi plants under dehydration

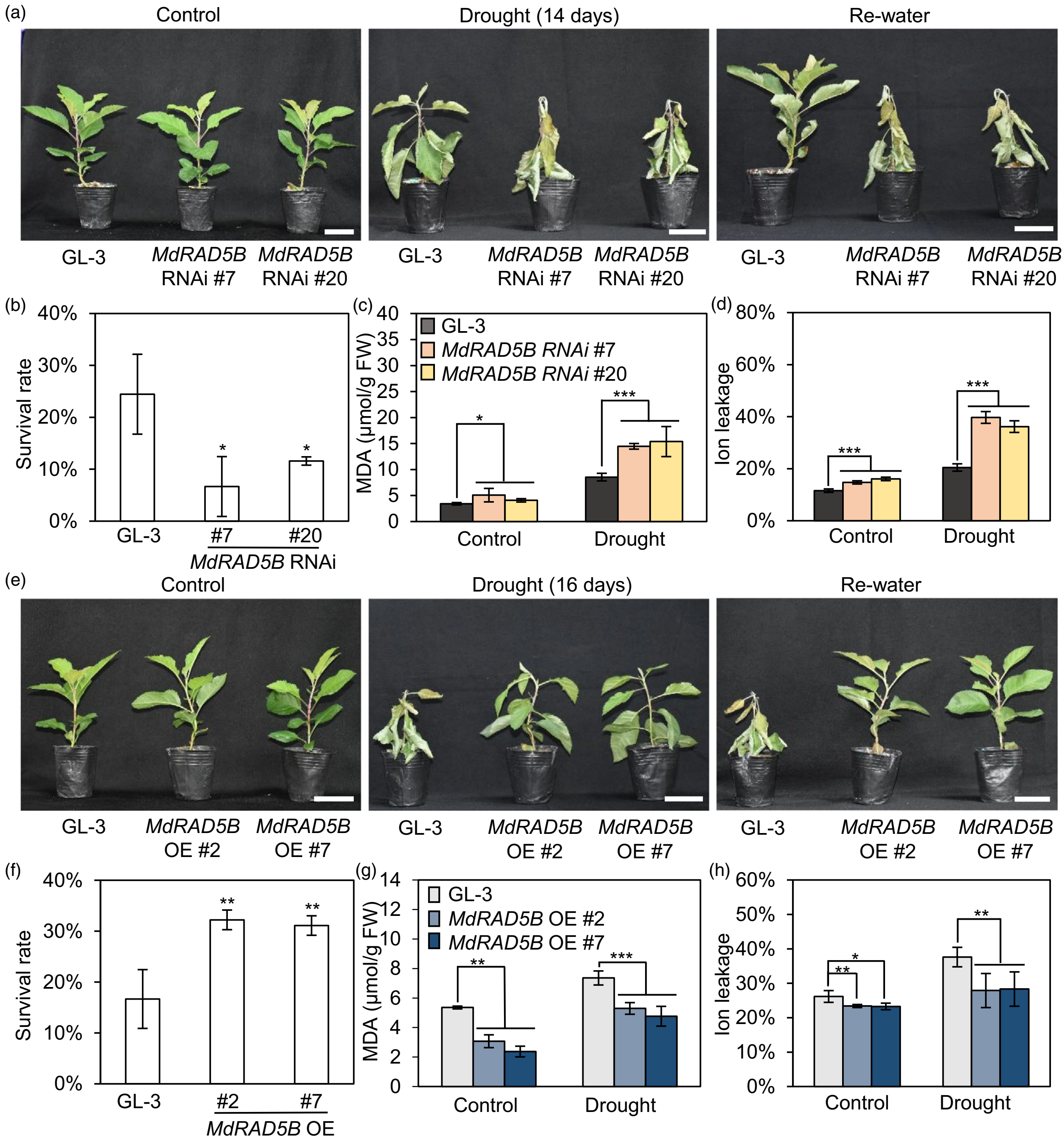


Figure 2 MdRAD5B plays a positive role in drought stress tolerance. (a) Whole-plant drought tolerance of *MdRAD5B* RNAi and GL-3 plants. Plants were withheld with water for 14 days and then re-water for 7 days. Bars = 5 cm. (b) Survival rate of *MdRAD5B* RNAi and GL-3 plants shown in (a). (c, d) Leaf MDA content and ion leakage of *MdRAD5B* RNAi and GL-3 plants after short-term drought treatment for 10 days. (e) Whole-plant drought tolerance of *MdRAD5B* OE and GL-3 plants. Plants were withheld with water for 16 days and then re-water for 7 days. Bars = 5 cm. (f) Survival rate of *MdRAD5B* OE and GL-3 plants shown in (e). (g, h) Leaf MDA content and ion leakage of *MdRAD5B* OE and GL-3 plants after drought stress for 12 days. Error bars indicate standard deviation [n = 27 in (a–c, e–g), 27 plants were used and each 9 plants was a biological replication; n = 9 in (d, h), 9 plants and three leaf discs from each plant were selected from control or drought treatment]. Student's t test was performed, and statistically significant differences were indicated by *P < 0.05, **P < 0.01, or ***P < 0.001.

conditions, whereas 945 drought-induced genes were down-regulated (Figure 3a–c; Data S3 and S4), indicating that MdRAD5B may enhance drought tolerance by regulating these drought-responsive genes.

We selected 10 genes, including *MdABF2-Like*, *MdABF3*, *MaATG8a*, *MdGBF3*, *MdHB7-Like*, *MdLEA14*, *MdSnRK2.6*, *MdTT7*, *MdCIPK11*, and *MdUGT71B6* to verify our RNA-seq data using qRT-PCR analysis. Among these 10 genes, homologues of *MdABF2-Like*, *MdABF3*, and *MdSnRK2.6* are involved in the ABA signalling pathway and can enhance drought tolerance (Fujita *et al.*, 2009; Kang *et al.*, 2002; Kim *et al.*, 2004). Homologues of *MdATG8a*, *MdGBF3*, *MdLEA14*, *MdHB7-Like*,

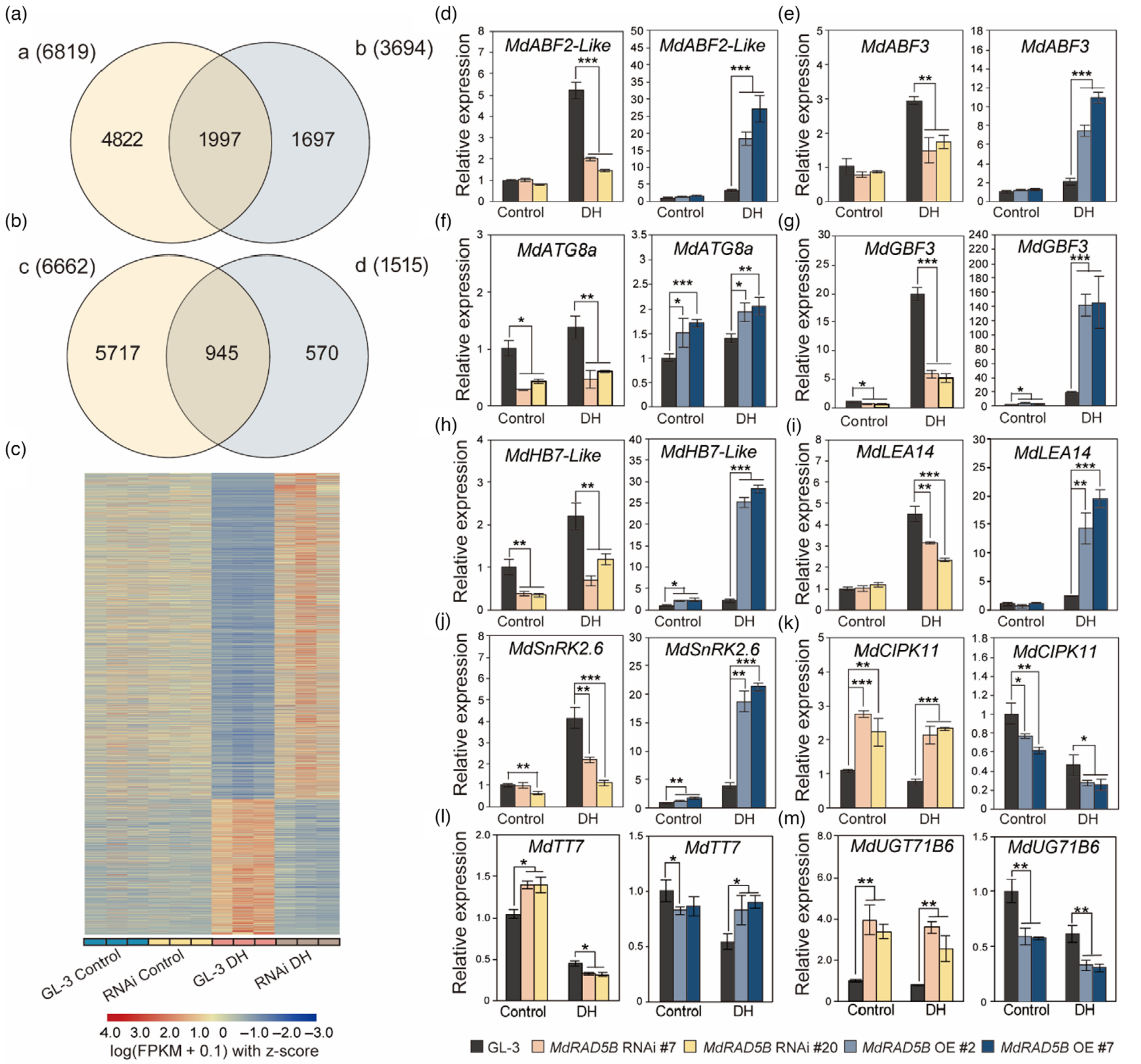


Figure 3 Regulation of drought-responsive genes by MdRAD5B in response to dehydration. (a) Venn diagrams of down-regulated genes in GL-3 plants under dehydration conditions (a) and differentially expressed genes (DEGs) which were up-regulated in *MdRAD5B* RNAi plants under dehydration conditions when compared with GL-3 plants (b). (b) Venn diagrams of up-regulated genes in GL-3 plants under dehydration (c) and differentially expressed genes (DEGs) which were down-regulated in *MdRAD5B* RNAi plants under dehydration when compared with GL-3 plants (d). The absolute value of $\log_2$FC for the significantly differentially expressed genes in the RNA-Seq portion of the Venn diagram is >1. (c) Heat map of the overlapped genes shown in (a) and (b). DH, dehydration; RNAi, *MdRAD5B* RNAi #20. (d–m) Validation of RNA-seq data shown in (a) and (b) using RT-qPCR. Error bars indicate standard deviation (n = 3). Student's t test was performed, and statistically significant differences were indicated by $*P < 0.05$, $**P < 0.01$, or $***P < 0.001$.

and *MdTT7* act as positive regulators of drought tolerance (Jia *et al*., 2021; Park *et al*., 2011; Ramegowda *et al*., 2017; Rao *et al*., 2020; Ré *et al*., 2014). However, homologues of *MdCIPK11* and *MdUGT71B6* negatively regulate drought tolerance (Dong *et al*., 2014; Ma *et al*., 2019). Compared with GL-3 plants under dehydration conditions, the expression of *MdABF2-Like*, *MdABF3*, *MdATG8a*, *MdGBF3*, *MdHB7-Like*, *MdLEA14*, *MdSnRK2.6*, and *MdTT7* was decreased in *MdRAD5B* RNAi plants and increased in *MdRAD5B* OE plants (Figure 3d–j,l). In contrast, the expression of *MdCIPK11* and *MdUGT71B6* was increased in *MdRAD5B* RNAi plants and decreased in *MdRAD5B* OE plants (Figure 3k,m). These results were consistent with the RNA-seq results, suggesting the reliability of the RNA-seq data.

MdRAD5B regulates chromatin remodelling of drought-responsive genes

Sequence alignment showed that MdRAD5B belongs to the SNF2 family, which acts as the major catalytic subunits of chromatin remodelling complexes (Figure S1). SNF2 proteins can use the energy of ATP hydrolysis to change the chromatin environment (Kang *et al*., 2022). Since the DEXDc domain of SNF2 family proteins plays a critical role in chromatin remodelling function

(Knizewski *et al*., 2008), we then purified recombinant MdRAD5B protein containing the DEXDc domain (371 aa to 800 aa) from *E.Coli a*nd determined its ATPase activity. According to the previous study (Sung *et al*., 1988), K562, a potential critical site for MdRAD5B ATPase activity (Figure S1), was mutated to A562 and served as the dead form of MdRAD5B. Results revealed that DNA could stimulate the ATPase activity of the MdRAD5B protein (371 aa to 800 aa) (Figure 4a). When DNA was absent or when MdRAD5B K562A was used, the ATPase activity of MdRAD5B was abolished (Figure 4a). These results suggest that MdRAD5B has a ATPase activity.

The stimulation of the ATPase activity of MdRAD5B by DNA suggested that MdRAD5B could be a chromatin remodelling factor. We used the assay for transposase-accessible chromatin with high-throughput sequencing (ATAC-seq) to examine the regulation of chromatin accessibility by MdRAD5B under control and dehydration conditions (Figures S4 and S5; Data S5). ATAC-seq analysis revealed a total of 17 010 differentially accessible regions (DARs) in GL-3 plants in response to dehydration (Data S6). 52.71% of these DARs was located in the promoter regions (−2000 to +100 bp from a TSS) (Figure S6a). ATAC-seq analysis also revealed a total of 6436 and 7369 DARs, which were substantially enriched in the promoter regions in *MdRAD5B* RNAi plants under control and dehydration conditions as compared with GL-3 plants, respectively (Figure 4b, Data S7 and S8). Among the 7369 DARs under dehydration conditions, 2025 DARs were increased while 5344 DARs were decreased, as compared with GL-3 plants (Data S8). These data indicate that MdRAD5B can regulate chromatin accessibility in response to drought stress.

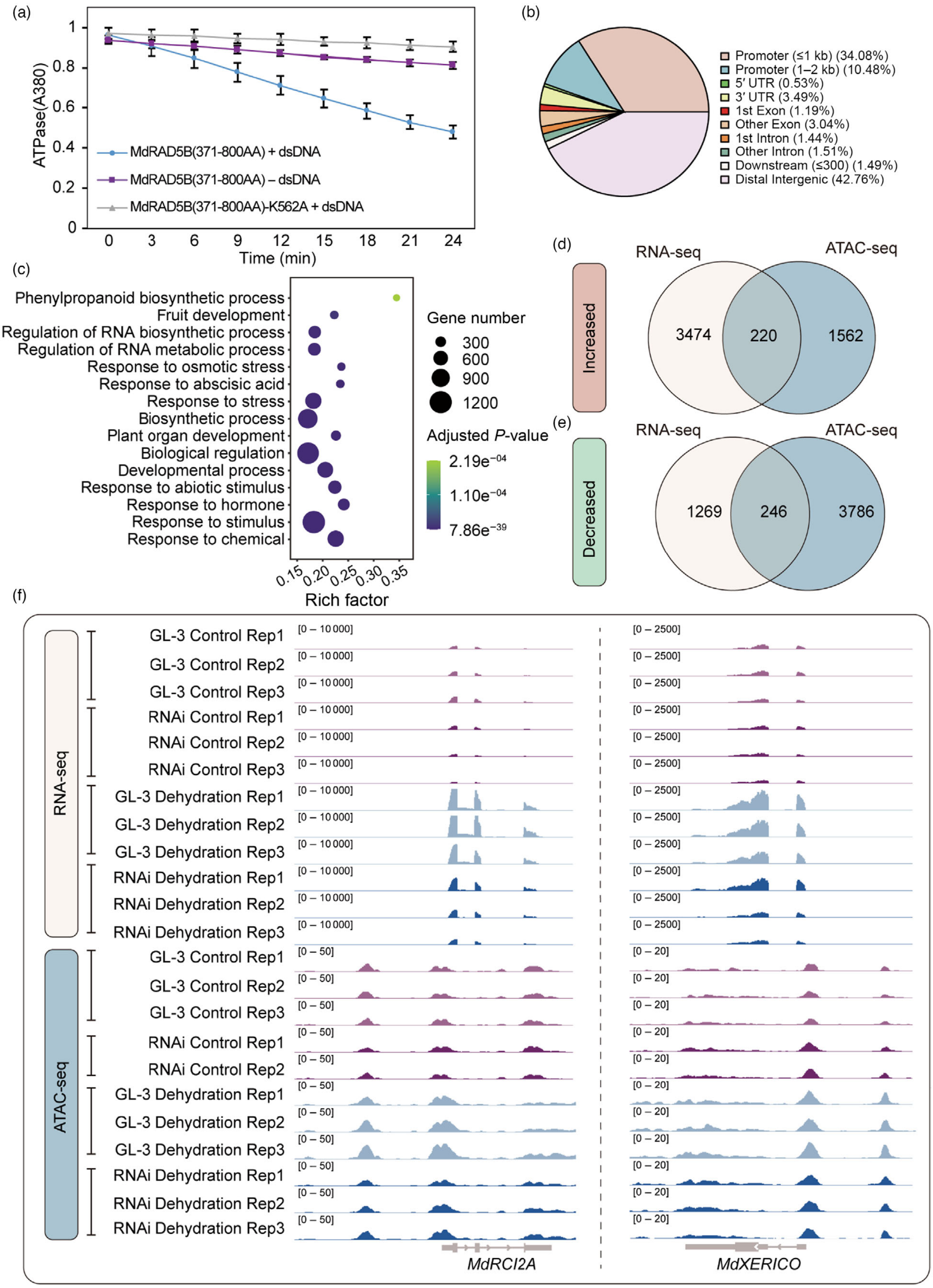
Since chromatin accessibility is associated with gene expression (Potter *et al*., 2018), we next determined whether the altered chromatin regions occurred near the genes that were transcriptionally regulated by MdRAD5B. We annotated each DAR to the closest gene using Chipseeker as most regulatory regions near the genes that they control (Sullivan *et al*., 2015). Because multiple DARs can be annotated to one gene, 17 010 DARs responding to dehydration in GL-3 plants were annotated to 11 910 genes that were enriched in response to abiotic stress and the plant development process (Figure S6b; Data S6). A combination of ATAC-seq and RNA-seq analysis revealed that altered chromatin accessibility affected more than 40% of the induced DEGs and 6.76% of the reduced DEGs by dehydration in GL-3 plants (Figure S6c,d), indicating that the altered chromatin accessibility may contribute to the altered gene expression under dehydration. In addition, 7369 DARs identified in *MdRAD5B* RNAi transgenic plants under dehydration conditions, as compared with GL-3 plants, were annotated to 5815 genes (Data S8). These 5815 genes were identified to participate in hormone response, abiotic stress, and the development process (Figure 4c). ATAC-seq and RNA-seq analysis revealed that 220 up-regulated DEGs in *MdRAD5B* RNAi transgenic plants had increased DARs under dehydration conditions, as compared with GL-3 plants (Figure 4d). On the other hand, the expression of 246 DEGs with decreased DARs was repressed in *MdRAD5B* RNAi transgenic plants under dehydration conditions as compared with GL-3 plants (Figure 4e). Among these 466 DEGs, homologues of *MdRCI2A*, *MdNF-YB3*, *MdXERICO*, *MdGBF3*, and *MdABF2-Like* are positive regulators of drought tolerance (Kim *et al*., 2004; Medina *et al*., 2005; Ramegowda *et al*., 2017; Sato *et al*., 2019; Vonapartis *et al*., 2022). The homologue of *MdTSPO* increases drought tolerance by enhancing ABA sensitivity (Guillaumot *et al*., 2009), while the homologue of *MdGIM2* decreases drought tolerance by negatively regulating ABA sensitivity (Liu *et al*., 2019). In *MdRAD5B* RNAi plants under dehydration conditions, we found that the expression of *MdRCI2A*, *MdXERICO*, *MdNF-YB3*, *MdTSPO*, *MdABF2-Like*, and *MdGBF3* was repressed. At the same time, the chromatin accessibility of these genes was decreased (Figure 4f; Figure S7). On the contrary, the expression of *MdGIM2* was enhanced and its chromatin accessibility was increased in *MdRAD5B* RNAi plants under dehydration conditions, as compared with GL-3 plants (Figure S7). These data suggest that MdRAD5B regulates the expression of drought-responsive genes under drought by modulating their chromatin accessibility, at least in part.

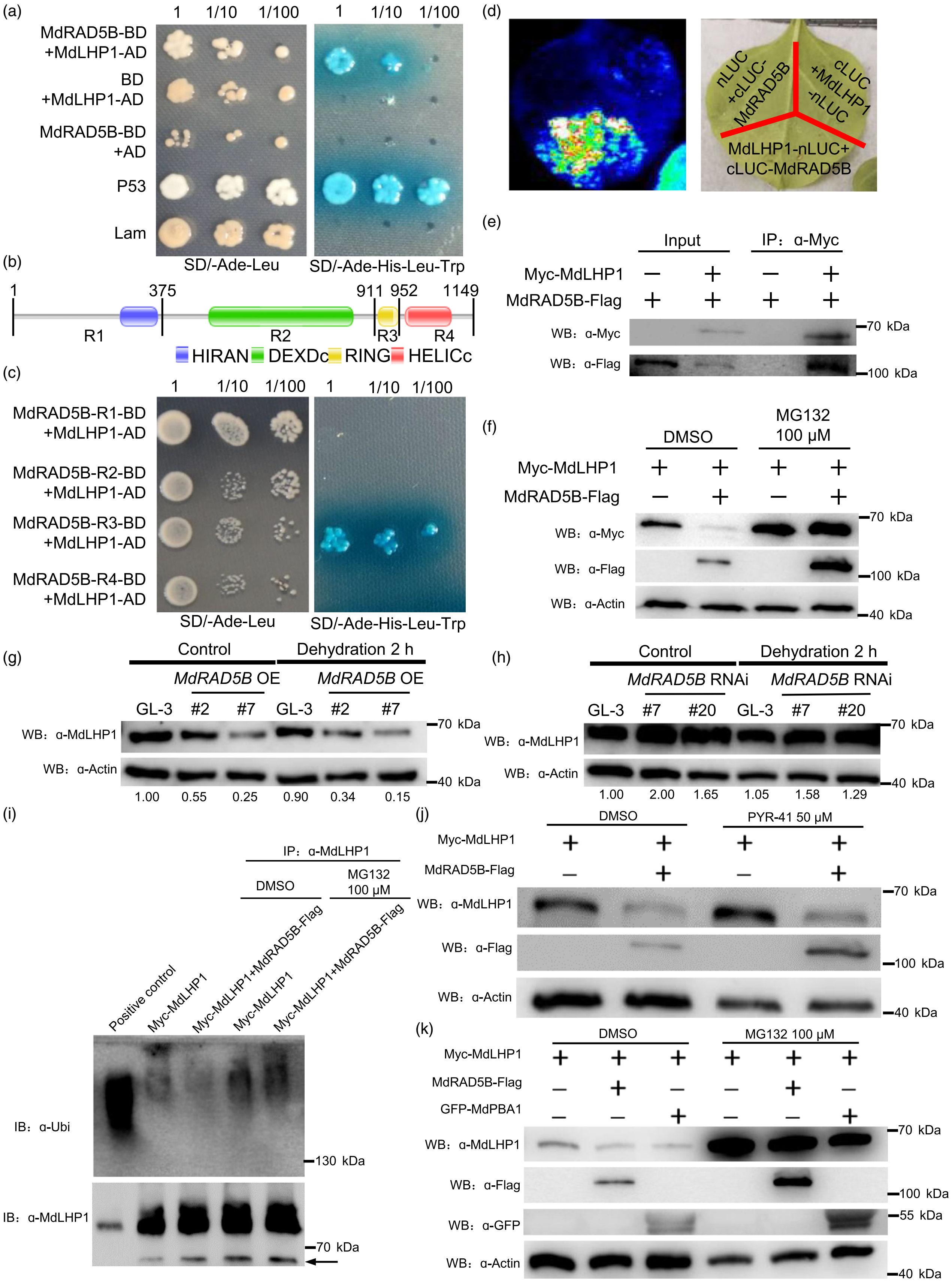
MdRAD5B interacts with and degrades MdLHP1

To further explore the regulatory mechanism of MdRAD5B, we performed a Yeast-two hybrid (Y2H) screen analysis. We identified several potential interacting proteins of MdRAD5B (Table S1), including a homologue of Arabidopsis LHP1, which is the "reader" of histone H3 lys27 (H3K27) trimethylation (Turck *et al*., 2007). Arabidopsis LHP1 is a negative regulator of drought tolerance (Ramirez-Prado *et al*., 2019). Y2H analysis showed that the RING domain of MdRAD5B was responsible for the interaction with MdLHP1 (Figure 5a–c). In addition, the Split-luciferase complementation (Split-LUC) assay and co-immunoprecipitation (Co-IP) further illustrate the *in vivo* interaction (Figure 5d,e).

Since the RING domain is involved in protein degradation (Deshaies and Joazeiro, 2009), we were interested in whether MdRAD5B could mediate the stability of MdLHP1. When Myc-MdLHP1 and MdRAD5B-Flag were co-expressed in tobacco leaves, the amount of Myc-MdLHP1 largely decreased (Figure 5f), suggesting that the degradation of MdLHP1 by MdRAD5B occurred. Moreover, this degradation was inhibited when the proteasome inhibitor, MG132, was applied (Figure 5f), indicating that MdRAD5B may degrade MdLHP1 *via* the 26S proteasome. In addition, the MdLHP1 protein level in *MdRAD5B* OE transgenic plants was significantly lower than in GL-3 plants

Figure 4 MdRAD5B regulates chromatin accessibility in response to dehydration. (a) ATPase activity of MdRAD5B. ATPase activity was measured spectrophotometrically with or without dsDNA. MdRAD5B K562A, a mutant of MdRAD5B with its Lys at 562 mutated to Ala. Error bars indicate standard deviation (n = 3). (b) Genomic distributions of the DARs in *MdRAD5B* RNAi plants under dehydration conditions compared to GL-3. (c) GO enrichment analysis of the genes adjacent to DARs, which were differentially regulated in *MdRAD5B* RNAi plants under dehydration conditions when compared with GL-3 plants. (d, e) Venn diagram showing the overlapped genes identified from RNA-seq and ATAC-seq in *MdRAD5B* RNAi plants compared with GL-3 plants under dehydration conditions. Gene and peak annotations were classified as increased (d) or decreased (e) according to the gene expression and chromatin accessibility change. The absolute value of $\log_2$FC for the significantly differentially expressed genes (P-adjust < 0.01) in the RNA-seq of the Venn diagram is >1. (f) Integrative Genomics Viewer (IGV) tracks showing examples of significantly changed genes at chromatin accessibility and transcription level in *MdRAD5B* RNAi plants under dehydration conditions, as compared with GL-3 plants. RNAi, *MdRAD5B* RNAi #20. DAR, differentially accessible region. Rep, replicate.


(a)
ATPase(A380)
Time (min)
MdRAD5B(371-800AA) + dsDNA
MdRAD5B(371-800AA) – dsDNA
MdRAD5B(371-800AA)-K562A + dsDNA
(b)
Promoter (≤1 kb) (34.08%)
Promoter (1–2 kb) (10.48%)
5′ UTR (0.53%)
3′ UTR (3.49%)
1st Exon (1.19%)
Other Exon (3.04%)
1st Intron (1.44%)
Other Intron (1.51%)
Downstream (≤300) (1.49%)
Distal Intergenic (42.76%)
(c)
Phenylpropanoid biosynthetic process
Fruit development
Regulation of RNA biosynthetic process
Regulation of RNA metabolic process
Response to osmotic stress
Response to abscisic acid
Response to stress
Biosynthetic process
Plant organ development
Biological regulation
Developmental process
Response to abiotic stimulus
Response to hormone
Response to stimulus
Response to chemical
Rich factor
Gene number
Adjusted P-value
(d)
Increased
RNA-seq
ATAC-seq
3474
220
1562
(e)
Decreased
RNA-seq
ATAC-seq
1269
246
3786
(f)
RNA-seq
ATAC-seq
GL-3 Control Rep1
GL-3 Control Rep2
GL-3 Control Rep3
RNAi Control Rep1
RNAi Control Rep2
RNAi Control Rep3
GL-3 Dehydration Rep1
GL-3 Dehydration Rep2
GL-3 Dehydration Rep3
RNAi Dehydration Rep1
RNAi Dehydration Rep2
RNAi Dehydration Rep3
MdRCI2A
MdXERICO



(a)
1 1/10 1/100 1 1/10 1/100
MdRAD5B-BD +MdLHP1-AD
BD +MdLHP1-AD
MdRAD5B-BD +AD
P53
Lam
SD/-Ade-Leu
SD/-Ade-His-Leu-Trp
(b)
1 375 911 952 1149
R1 R2 R3 R4
HIRAN DEXDc RING HELICc
(c)
1 1/10 1/100 1 1/10 1/100
MdRAD5B-R1-BD +MdLHP1-AD
MdRAD5B-R2-BD +MdLHP1-AD
MdRAD5B-R3-BD +MdLHP1-AD
MdRAD5B-R4-BD +MdLHP1-AD
SD/-Ade-Leu
SD/-Ade-His-Leu-Trp
(d)
nLUC +cLUC-MdRAD5B
cLUC +MdLHP1 -nLUC
MdLHP1-nLUC+ cLUC-MdRAD5B
(e)
Input
IP：α-Myc
Myc-MdLHP1 − + − +
MdRAD5B-Flag + + + +
WB：α-Myc
WB：α-Flag
70 kDa
100 kDa
(f)
DMSO
MG132 100 μM
Myc-MdLHP1 + + + +
MdRAD5B-Flag − + − +
WB：α-Myc
WB：α-Flag
WB：α-Actin
70 kDa
100 kDa
40 kDa
(g)
Control
Dehydration 2 h
MdRAD5B OE
MdRAD5B OE
GL-3 #2 #7 GL-3 #2 #7
WB：α-MdLHP1
WB：α-Actin
70 kDa
40 kDa
1.00 0.55 0.25 0.90 0.34 0.15
(h)
Control
Dehydration 2 h
MdRAD5B RNAi
MdRAD5B RNAi
GL-3 #7 #20 GL-3 #7 #20
WB：α-MdLHP1
WB：α-Actin
70 kDa
40 kDa
1.00 2.00 1.65 1.05 1.58 1.29
(i)
IP：α-MdLHP1
DMSO
MG132 100 μM
Positive control
Myc-MdLHP1
Myc-MdLHP1+MdRAD5B-Flag
Myc-MdLHP1
Myc-MdLHP1+MdRAD5B-Flag
IB：α-Ubi
IB：α-MdLHP1
130 kDa
70 kDa
(j)
DMSO
PYR-41 50 μM
Myc-MdLHP1 + + + +
MdRAD5B-Flag − + − +
WB：α-MdLHP1
WB：α-Flag
WB：α-Actin
70 kDa
100 kDa
40 kDa
(k)
DMSO
MG132 100 μM
Myc-MdLHP1 + + + + + +
MdRAD5B-Flag − + − − + −
GFP-MdPBA1 − − + − − +
WB：α-MdLHP1
WB：α-Flag
WB：α-GFP
WB：α-Actin
70 kDa
100 kDa
55 kDa
40 kDa

Figure 5 MdRAD5B interacts with and degrades MdLHP1 *via* ubiquitin-independent 20S proteasome. (a) Interaction of MdRAD5B and MdLHP1 determined by yeast-two hybrid (Y2H) assays. The full-length MdRAD5B protein was fused with the GAL4-binding domain, and the full-length MdLHP1 protein was fused with the GAL4 activation domain. P53 was used as a positive control, and Lam was used as a negative control. (b) Domain structure of MdRAD5B. MdRAD5B was divided into four regions and used for Y2H assays. R, region. R1, 1–375 aa. R2, 376–911 aa. R3, 912–951 aa. R4, 952–1149 aa. (c) Interaction of MdRAD5B deletions and MdLHP1 determined by Y2H assays. The MdRAD5B deletions were fused with the GAL4-binding domain and the full-length MdLHP1 protein was fused with the GAL4 activation domain. (d) Luciferase complementation imaging assays showing the interaction between MdRAD5B and MdLHP1 in tobacco (*Nicotiana tabacum*). MdLHP1 was fused with nLUC (N terminus of LUC) and MdRAD5B was fused with cLUC (C terminus of LUC). The LUC signals were observed after infiltration for 3 days. (e) Interaction of MdRAD5B and MdLHP1 determined by Co-IP assay. (f) The degradation of MdLHP1 by MdRAD5B in tobacco. For MG132 treatment, leaves were treated with DMSO or 100 μM MG132 for 12 h after 2 days of infiltration. (g, h) MdLHP1 levels in GL-3, *MdRAD5B* OE (g), and *MdRAD5B* RNAi (h) plants under control and dehydration conditions. OE, overexpression. (i) The ubiquitination of MdLHP1 by MdRAD5B under DMSO or 100 μM MG132 treatment. (j) The stability of MdLHP1 by MdRAD5B under DMSO and PYR-41 treatment. For PYR-41 treatment, leaves were treated with DMSO or 50 μM PYR-41 for 12 h after 2 days of infiltration. (k) The stability of MdLHP1 by MdRAD5B or MdPBA1 under DMSO or 100 μM MG132 treatment.

under control and dehydration conditions (Figure 5g). In *MdRAD5B* RNAi transgenic plants, on the contrary, the protein level of MdLHP1 was higher than that in GL-3 plants under control and dehydration conditions (Figure 5h). These results further suggest that MdRAD5B degrades MdLHP1 in response to drought and that this degradation might occur *via* the 26S proteasome pathway.

MdRAD5B mediates the MdLHP1 degradation *via* ubiquitin-independent 20S proteasome

Since MG132 is a potent proteasome inhibitor and ubiquitin tagging of a protein is often targeted to the 26S proteasome for irreversible degradation (Bard *et al*., 2018). Therefore, we asked whether MdRAD5B degraded MdLHP1 through the ubiquitin-dependent 26S proteasome pathway, using ubiqunitinated MdBBX7 as a positive control (Chen *et al*., 2022). Results showed that although slight increase of ubiquitinated Myc-MdLHP1 was detected after MG132 treatment, the ubiquitination of Myc-MdLHP1 was not due to MdRAD5B-Flag (Figure 5i), implying that MdRAD5B degrades MdLHP1 not through the ubiquitin-dependent 26S proteasome.

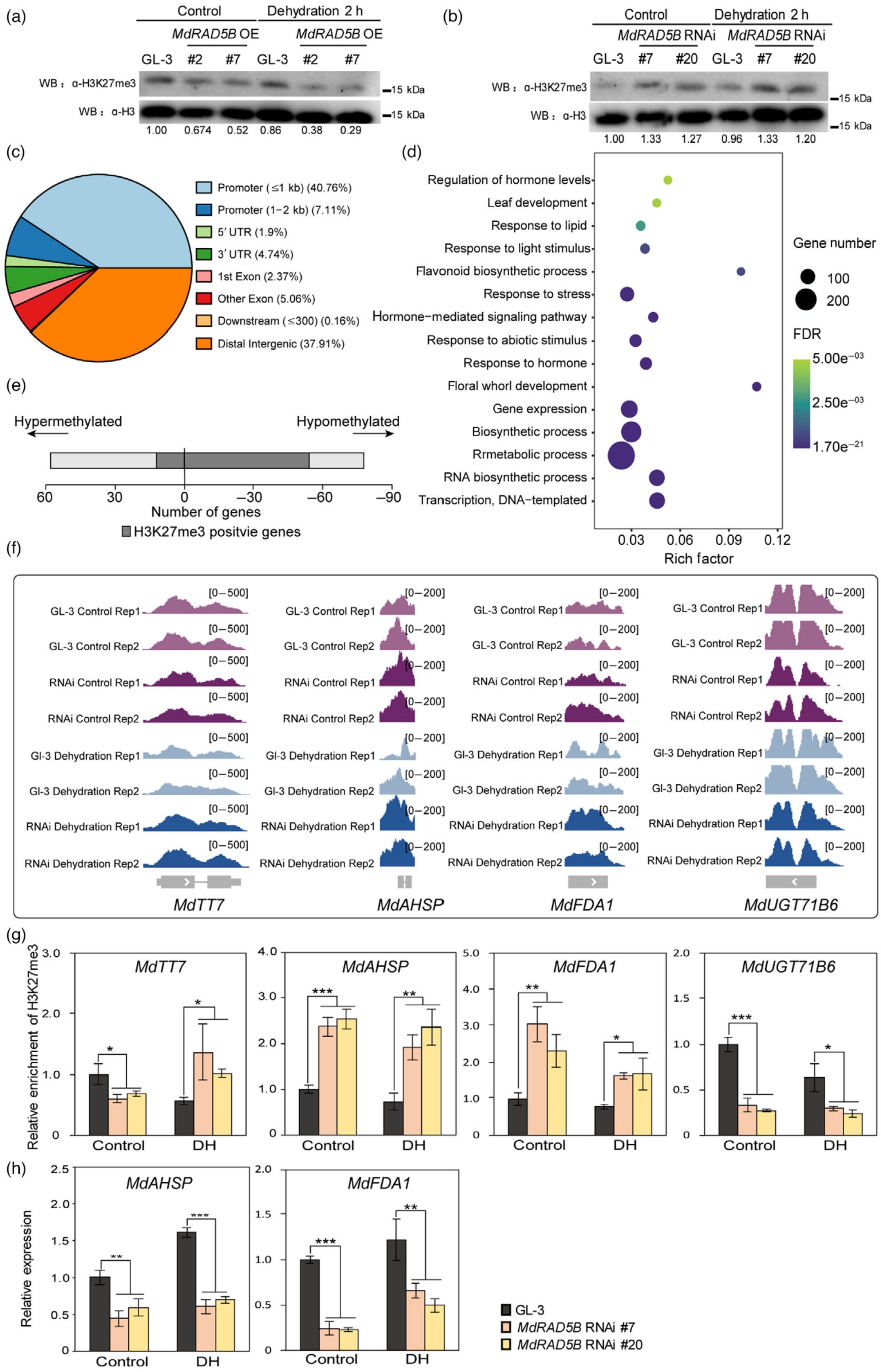
Besides the ubiquitin-dependent 26S proteasome pathway, protein degradation can also occur through the ubiquitin-independent 20S core proteasome pathway, which can be inhibited by MG132 (Li *et al*., 2020b). To investigate if the 20S core proteasome accounts for MdLHP1 degradation, we treated tobacco leaves expressing Myc-MdLHP1 and MdRAD5B-Flag with PYR-41, a protease inhibitor of ubiquitin-activating enzyme E1, to selectively impede the ubiquitin-dependent 26S proteasome pathway but not the ubiquitin-independent 20S proteasome pathway. As expected, when PYR-41 was absent, MdLHP1 was again degraded quickly by MdRAD5B (Figure 5j). However, the degradation of MdLHP1 by MdRAD5B was still not deterred by PYR-41 (Figure 5j), supporting the notion that MdRAD5B might degrade MdLHP1 through the ubiquitin-independent 20S core proteasome.

The 20S core particle proteasome (CP) hydrolyses the unfolded polypeptide into short peptides or amino acids (Bard *et al*., 2018). It is a self-segregating multicatalytic protease consisting of four stacked heptad rings in which the α and β subunits are assembled in $\alpha_{1\text{-}7}/\beta_{1\text{-}7}/\beta_{1\text{-}7}/\alpha_{1\text{-}7}$ conformation (Book *et al*., 2010). To understand which subunit is responsible for MdLHP1 degradation by MdRAD5B *via* the 20S proteasome, we cloned a few components of the 20S proteasome from *Malus domestica* (Figure S8a) and conducted Y2H and Split-LUC assays to examine the interaction between 20S proteasome components and MdLHP1 or MdRAD5B. The Split-LUC assay showed that MdLHP1 indeed interacted with MdPBA1 (β1) *in vivo*, but not with MdPAG1 (α7) and MdPBG1 (β7) (Figure S8b–d). In addition, MdRAD5B interacted with MdPBA1 *in vivo* as well (Figure S8e). We also verified the *in vivo* interaction of MdPBA1 with MdLHP1 or MdRAD5B by Co-IP assay (Figure S8f, g). However, neither MdRAD5B nor MdLHP1 interacted with MdPBA1 *in vitro*, as determined by Y2H analysis (Figure S8h). Furthermore, the degradation of MdLHP1 by MdPBA1 could be inhibited by MG132 (Figure 5k). These results indicate that the degradation of MdLHP1 by MdRAD5B might be *via* MdPBA1, a subunit of the 20S proteasome, in a ubiquitin-independent 20S proteasome way.

MdRAD5B modulates H3K27me3 deposition on drought-responsive genes in response to drought

Arabidopsis LHP1 is a subunit of PRC1 and interacts with MSI1 to recruit the PRC2 complex to maintain H3K27me3 histone modification (Derkacheva *et al*., 2013). As the homologue of AtLHP1, MdLHP1 was localized in the nucleus (Figure S9). In addition, MdLHP1 interacted with MdMSI1 *in vivo* (Figure S10). Further experiments showed that MdLHP1 can bind H3K27me3 and maintain the H3K27me3 level (Figure S11). These results indicate that MdLHP1 can regulate H3K27me3 deposition in *Malus domestica*, similar to its counterpart in Arabidopsis.

Because MdLHP1 regulated H3K27me3 deposition and could be degraded by MdRAD5B, we then asked if MdRAD5B could affect H3K27me3 deposition in response to drought stress. As shown in Figure 6a, the *MdRAD5B* OE plants showed a lower H3K27me3 level than GL-3 plants under control and dehydration conditions. The *MdRAD5B* RNAi plants, in contrast, exhibited a higher H3K27me3 level than GL-3 plants under control and dehydration conditions (Figure 6b). Furthermore, we used chromatin immunoprecipitation with sequencing (ChIP-seq) to examine the genome-wide regulation of H3K27me3 by MdRAD5B under control and dehydration conditions (Figures S12 and S13; Data S9). ChIP-seq analysis of GL-3 plants between control and dehydration conditions revealed a total of 3445 differentially methylated (H3K27me3) regions ($|\log_2 FC| > 0.5$, *P*-value <0.01), which were annotated to 3123 genes enriched in plant development or response to abiotic stress and hormones (Data S10; Figure S14a). Among the 3123 differentially methylated genes, expression of 1268 genes was identified to significantly up- or down-regulated (*P*-adjust <0.05) under dehydration conditions in GL-3 plants (Figure S14b; Data S4 and S10). ChIP-seq analysis also revealed a total of 2244 and 633



(a)
Control
Dehydration 2 h
MdRAD5B OE
MdRAD5B OE
GL-3 #2 #7 GL-3 #2 #7
WB：α-H3K27me3
WB：α-H3
15 kDa
1.00 0.674 0.52 0.86 0.38 0.29
(b)
Control
Dehydration 2 h
MdRAD5B RNAi
MdRAD5B RNAi
GL-3 #7 #20 GL-3 #7 #20
1.00 1.33 1.27 0.96 1.33 1.20
(c)
Promoter (≤1 kb) (40.76%)
Promoter (1−2 kb) (7.11%)
5′ UTR (1.9%)
3′ UTR (4.74%)
1st Exon (2.37%)
Other Exon (5.06%)
Downstream (≤300) (0.16%)
Distal Intergenic (37.91%)
(d)
Regulation of hormone levels
Leaf development
Response to lipid
Response to light stimulus
Flavonoid biosynthetic process
Response to stress
Hormone−mediated signaling pathway
Response to abiotic stimulus
Response to hormone
Floral whorl development
Gene expression
Biosynthetic process
Rrmetabolic process
RNA biosynthetic process
Transcription, DNA−templated
Rich factor
Gene number
100
200
FDR
5.00e−03
2.50e−03
1.70e−21
(e)
Hypermethylated
Hypomethylated
Number of genes
H3K27me3 positvie genes
(f)
GL-3 Control Rep1
GL-3 Control Rep2
RNAi Control Rep1
RNAi Control Rep2
Gl-3 Dehydration Rep1
Gl-3 Dehydration Rep2
RNAi Dehydration Rep1
RNAi Dehydration Rep2
[0−500]
[0−200]
MdTT7
MdAHSP
MdFDA1
MdUGT71B6
(g)
Relative enrichment of H3K27me3
Control
DH
(h)
Relative expression
GL-3
MdRAD5B RNAi #7
MdRAD5B RNAi #20

Figure 6 MdRAD5B regulates H3K27me3 deposition in response to dehydration. (a, b) Endogenous H3K27me3 level in GL-3, *MdRAD5B* OE (a), and *MdRAD5B* RNAi (b) plants under control and dehydration conditions. OE, overexpression. (c) A pie chart showing the genomic distribution of differentially methylated regions identified in *MdRAD5B* RNAi plants compared to GL-3 under dehydration conditions. (d) GO enrichment analysis of differentially methylated (H3K27me3) genes between GL-3 and *MdRAD5B* RNAi plants under dehydration conditions. (e) Number of DEGs (*P*-value <0.05) and H3K27me3 positive genes in *MdRAD5B* RNAi plants under dehydration conditions, as compared with GL-3 plants. H3K27me3-positive genes include the DEGs with up-regulated expression and down-regulated H3K27me3 or down-regulated expression and up-regulated H3K27me3 level. (f) Integrative Genomics Viewer (IGV) tracks showing examples of significantly altered genes at H3K27me3 abundance and transcription level in *MdRAD5B* RNAi plants under dehydration conditions, as compared with GL-3 plants. (g) Validation of ChIP-seq data shown in (f) using ChIP-qPCR. (h) Validation of RNA-seq data shown in (f) using RT-qPCR. RNAi, *MdRAD5B* RNAi #20. DH, dehydration. Rep, replicate. Error bars indicate standard deviation (n = 3). Student's *t* test was performed and statistically significant differences were indicated by $*P < 0.05$, $**P < 0.01$, or $***P < 0.001$.

differentially methylated (H3K27me3) regions ($|\log_2 FC| > 0.5$, *P*-value <0.01) in *MdRAD5B* RNAi plants under control and dehydration conditions, respectively, as compared with GL-3 plants (Data S11 and S12). Under dehydration conditions, the 633 differentially methylated (H3K27me3) regions, which were substantially located in the promoter region, were annotated to 615 genes (Figure 6c). GO enrichment analysis showed that these 615 genes were mainly enriched in metabolic process and response to abiotic stress and hormone (Figure 6d). Among these 615 differentially methylated (H3K27me3) genes under dehydration conditions, 261 genes were hyper-methylated and 354 genes were hypo-methylated (Data S12). In addition, 66 differentially methylated genes were among the DEGs whose expression levels were altered (*P*-adjust <0.05) in *MdRAD5B* RNAi transgenic plants under dehydration conditions, as compared with GL-3 plants (Figure 6e; Data S3 and S12). These results suggest that MdRAD5B can regulate H3K27me3 deposition in response to drought stress.

We selected six genes to verify our ChIP-seq data using ChIP-qPCR analysis. Among these genes, the homologue of *MdTT7* acts as a positive regulator of drought tolerance in common bean (Leitão *et al*., 2021). In Aribidopsis, the homologue of *MdHNL13*, which has a conserved LEA-14 like domain, promotes ABA biogenesis under osmotic stress (Bao *et al*., 2016). Overexpression of *UGT71B6*, the homologue of *MdUGT71B6*, decreases drought tolerance of Arabidopsis (Dong *et al*., 2014), and *IAA16*, which is the homologue of *MdIAA16*, decreases the ABA sensitivity in Arabidopsis (Rinaldi *et al*., 2012). The Integrative Genomics Viewer (IGV) tracks and ChIP-qPCR results showed that the four hyper-methylated genes showed increased H3K27me3 deposition and the two hypo-methylated genes showed decreased H3K27me3 deposition in *MdRAD5B* RNAi transgenic plants compared with GL-3 plants under dehydration conditions, suggesting the reliability of the ChIP-seq data (Figure 6f,g; Figures S15 and S16a). We also examined the transcription level of these six differentially methylated genes under control and dehydration conditions in *MdRAD5B* RNAi and GL-3 plants. Results showed that expression of the four hyper-methylated genes was decreased in *MdRAD5B* RNAi transgenic plants under dehydration conditions as compared with GL-3 plants, while expression of the two hypo-methylated genes was increased (Figures 3l,m and 6h; Figure S16b). On the contrary, *MdRAD5B* OE plants displayed opposite expression levels and H3K27me3 levels of the detected genes to *MdRAD5B* RNAi plants under dehydration conditions. (Figure 3l,m; Figures S17 and S18). These results revealed that MdRAD5B can regulate the expression of drought-responsive genes by regulating H3K27me3 deposition in response to drought stress.

MdRAD5B acts upstream of MdLHP1

To investigate the genetic interaction of MdRAD5B and MdLHP1, we generated transgenic calli expressing reduced *MdRAD5B* (*MdRAD5B* RNAi) as well as double transgenic calli with reduced *MdRAD5B* and *MdLHP1* (*MdLHP1* anti/*MdRAD5B* RNAi) (Figures S19 and S20). We then treated the wild type and transgenic calli with PEG8000 to simulate low water potential (−0.7 MPa). After PEG treatment for 14 days, the relative growth rate of *MdLHP1* anti calli was higher than that of the wild type, while *MdRAD5B* RNAi calli had a lower relative growth rate than the wild type (Figure 7a,b; Figure S21). In addition, reduced expression of *MdLHP1* could recover the decreased relative growth rate of *MdRAD5B* RNAi calli under PEG treatment (Figure 7a,b; Figure S21), indicating that MdLHP1 acts downstream of MdRAD5B in drought stress response.

We also investigated gene expression and H3K27me3 deposition in the wild type and transgenic calli in response to PEG treatment. Under PEG treatment, the enrichment of H3K27me3 in drought-responsive genes including *MdTT7*, *MdNHL13*, and *MdAHSP* was greater in *MdRAD5B* RNAi calli but less in *MdLHP1* anti calli, as compared to the wild type. However, the increased H3K27me3 enrichment of these genes in *MdRAD5B* RNAi calli was hampered by the decreased expression of *MdLHP1* under PEG treatment (Figure 7c). Consistently, expression of these genes was reduced in *MdRAD5B* RNAi calli but increased in *MdLHP1* anti calli, and the reduced expression of these genes in *MdRAD5B* RNAi calli was recovered by decreased expression of *MdLHP1*, under PEG treatment (Figure 7d). On the contrary, under PEG treatment, H3K27me3 level of *MdIAA16* was lower in *MdRAD5B* RNAi calli, but comparable in *MdLHP1* anti and double transgenic calli, as compared with the wild type. Consistently, *MdRAD5B* RNAi calli showed increased *MdIAA16* expression level, but *MdLHP1* anti calli and double transgenic calli displayed comparable level, as compared with the wild type under PEG treatment (Figure 7c,d).These results further suggest that MdRAD5B functions upstream of MdLHP1 and modulates H3K27me3 deposition and expression of drought-responsive genes, including *MdTT7*, *MdNHL13*, *MdAHSP*, and *MdIAA16* through destabilizing MdLHP1 under drought stress, at least in part.

The RING and DEXDc domains are required for MdRAD5B-mediated drought tolerance

As shown in Figures 4 and 5, the DEXDc and RING domains were responsible for the ATPase activity and interaction with MdLHP1, respectively. Therefore, we next asked whether these two domains of MdRAD5B play roles in drought tolerance.

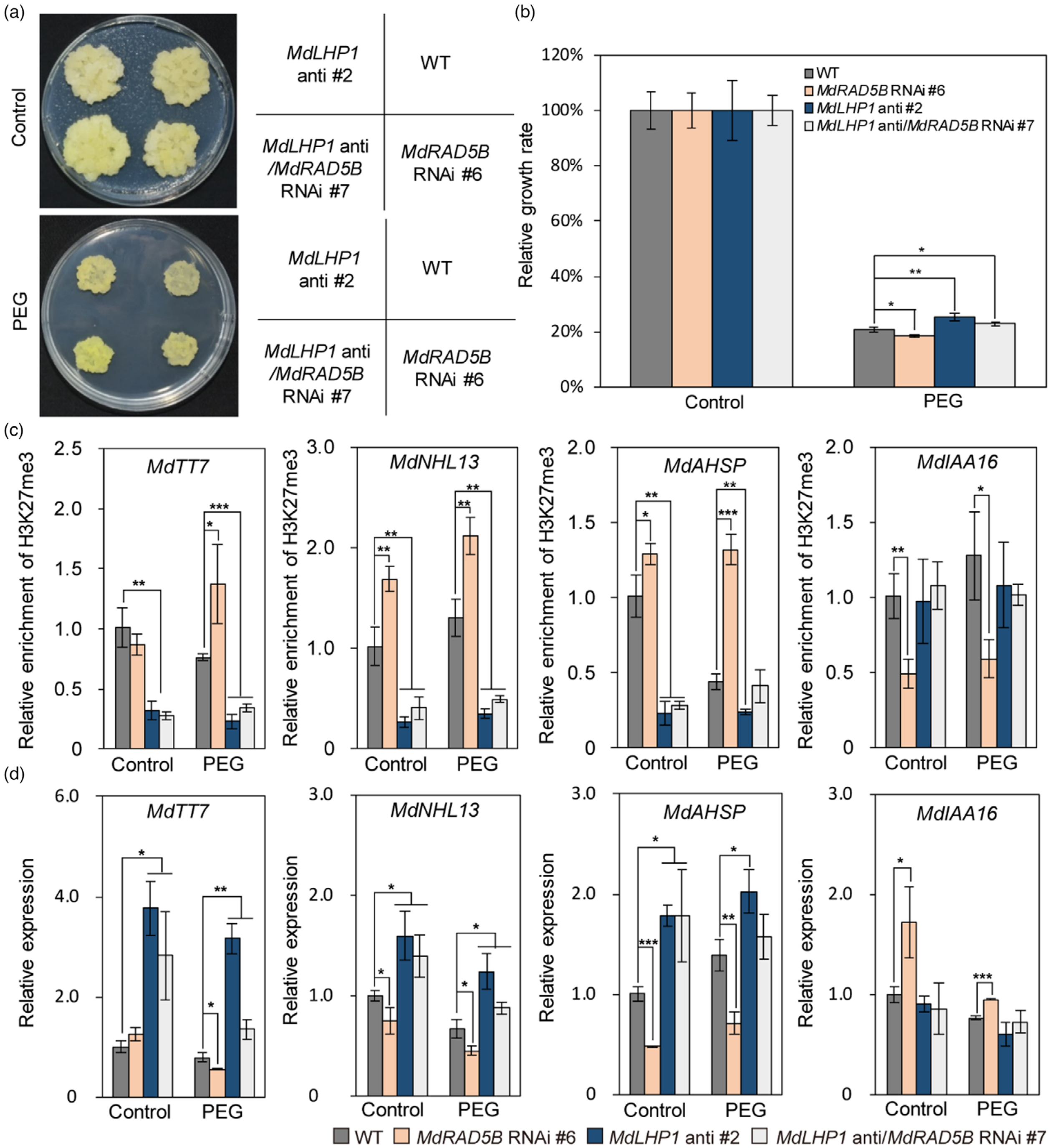


Figure 7 Genetic analysis of *MdRAD5B* and *MdLHP1*. (a) Morphological characteristics of the WT, *MdLHP1* anti, *MdRAD5B* RNAi, and *MdLHP1* anti/ *MdRAD5B* RNAi transgenic apple calli in response to PEG treatment. (b) The relative growth rate of WT, *MdLHP1* anti, *MdRAD5B* RNAi, and *MdLHP1* anti/ *MdRAD5B* RNAi transgenic apple calli under control or PEG treatment. (c, d) The H3K27me3 abundance (c) and transcription level (d) of drought-responsive genes in WT, *MdLHP1* anti, *MdRAD5B* RNAi, and *MdLHP1* anti/*MdRAD5B* RNAi transgenic apple calli under control or PEG treatment. *MdLHP1* anti/ *MdRAD5B* RNAi, knocking down *MdLHP1* in the background of *MdRAD5B* RNAi calli. Equal weight of WT and transgenic calli were grown on MS medium or MS medium supplemented with PEG (−0.7 MPa) for 2 weeks. WT, wild type. Error bars indicate standard deviation [$n = 5$ in (b), $n = 3$ in (c, d)]. Student's t test was performed and statistically significant differences were indicated by *$P < 0.05$, **$P < 0.01$, or ***$P < 0.001$.

We generated transgenic calli expressing *35S: MdRAD5B* (*MdRAD5B* OE), or *35S:MdRAD5B* without the RING domain (*MdRAD5B-△RING* OE), or *35S:MdRAD5B* without the DEXDc domain (*MdRAD5B-△DEXDc* OE) (Figure S22). We treated the wild type and transgenic calli with PEG for 14 days. After PEG treatment, *MdRAD5B* OE calli had a higher relative growth rate than the wild type. However, the transgenic *MdRAD5B-△DEXDc* OE calli had a lower relative growth rate as compared with the *MdRAD5B* OE calli, though they still had a greater relative growth rate than the wild type (Figure S23a,b), indicating the requirement of the DEXDc domain in drought tolerance of MdRAD5B. A similar pattern was found for the RING domain (Figure S23c,d). This data suggests that both the DEXDc and RING domains are required for the function of MdRAD5B in apple drought tolerance.

HIRAN domain of RAD5A is responsible for plant DNA repair (Iyer *et al*., 2006). However, RAD5B is reported to be dispensable for DNA repair (Iyer *et al*., 2006), indicating a distinct role for the HIRAN domain between RAD5A and RAD5B. We also generated the transgenic calli expressing *35S:MdRAD5B* without the HIRAN domain (*MdRAD5B-▵HIRAN* OE) (Figure S22). After PEG treatment, the relative growth rate of *MdRAD5B-▵HIRAN* OE calli was comparable with that of *MdRAD5B* OE calli, though it was greater than that of the wild type (Figure S23e,f), suggesting the unnecessity of the HIRAN domain of MdRAD5B in apple drought tolerance.

Discussion

As sessile plants, apple trees are frequently affected by drought (Liao *et al*., 2017). To breed drought-tolerant cultivars, characterization of drought-responsive genes, including epigenetic regulators, is inevitable to apply molecular breeding. Epigenetic regulators promote phenotypic plasticity in the development, survival, and reproduction of plants in adverse environments. These regulators include histone and DNA modifiers, histone variants, chromatin remodelers, and regulatory non-coding RNAs (Kim, 2021). Multiple sequence alignment and domain analysis revealed that MdRAD5B belongs to the SNF2 family (Figure S1). In general, SNF2 proteins use ATP to alter chromatin accessibility by remodelling chromatin structure (Clapier and Cairns, 2009; Ryan and Owen-Hughes, 2011). Up to now, 41 SNF2 proteins in Arabidopsis have been classified into several groups according to their protein structures (Knizewski *et al*., 2008). RAD5B belongs to the RAD5/16-like family, a subfamily of SNF2. The homologue of RAD5B, RAD5A, is a key regulator of DNA damage repair. However, RAD5B is dispensable for the DNA repair function (Chen *et al*., 2008). In this study, both ATPase activity and ATAC-seq analysis revealed that MdRAD5B does have a chromatin remodelling function (Figure 4a; Figures S4 and S5).

Our results showed that MdRAD5B is a positive regulator of drought tolerance in apple (Figure 2). However, we noticed that the MDA content and ion leakage of transgenic plants showed significant differences compared with GL-3 plants under control conditions (Figure 2c,d,g,h). We speculated that these basal responses were due to the altered expression of the MdRAD5B target genes under control conditions. ABA and ABA signalling can induce the formation of protective enzymes in the biomembrane system, reduce the degree of membrane lipid peroxidation, and protect the integrity of the membrane structure and photosynthetic characteristics (Chen *et al*., 2021; Tang *et al*., 2022; Wang *et al*., 2020; Zhu, 2016). Under control conditions, MdRAD5B positively regulates the expression of *MdSnRK2.6* (Figure 3j), whose homologue acts as a positive regulator in ABA signalling (Chen *et al*., 2020). Moreover, the expression of *MdUGT71B6* is negatively regulated by MdRAD5B under control conditions (Figure 3m). A homologue of *MdUGT71B6* is a negative regulator of ABA biogenesis (Dong *et al*., 2014). In summary, we hypothesized that the differential expression of such genes in *MdRAD5B* transgenic plants might lead to the basal response of MdRAD5B under control conditions.

Previous research showed that the sequence of Gly-X-Gly-Lys-Thr is crucial for the proteins that can bind and hydrolyze the purine nucleotides. Yeast RAD3 ATPase and DNA helicase activities are abolished when a point mutation of Lys-48, an invariant residue in the Gly-X-Gly-Lys-Thr sequence, is mutated to arginine (Sung *et al*., 1988). This sequence was also found in the DEXDc domain of MdRAD5B (Figure S1). The recombinant MdRAD5B protein containing the point mutation of Lys-562 to arginine within the Gly-X-Gly-Lys-Thr sequence in the DEXDc domain could abolish the ATPase activity of MdRAD5B (Figure 4a), indicating the functional conservation among RAD sub-family proteins in ATPase activity.

As one of the transcriptional regulation mechanisms, chromatin accessibility is thought to be context dependent (Asensi-Fabado *et al*., 2017). Previous work has shown that SNF2 family proteins regulate gene expression through their chromatin remodelling function (Clapier *et al*., 2017). BAF60 could bind to nucleosome-free regions of expressed genes and act antagonistically to PIF4 to regulate hypocotyl elongation by modulating DNA accessibility (Jégu *et al*., 2017). CHR19 impacts the transcription of genes to respond to different pathogens through its chromatin remodelling function (Kang *et al*., 2022). In our results, ATAC-seq and RNA-seq data showed that the expression of 466 DEGs was affected by changed chromatin accessibility in *MdRAD5B* RNAi transgenic plants compared with GL-3 plants under dehydration conditions (Figure 4d,e), suggesting that MdRAD5B might modulate the expression of drought-responsive genes through its chromatin remodelling function. However, many DEGs did not have nearby DARs in *MdRAD5B* RNAi plants under control and dehydration conditions, as compared with GL-3 plants (Figure 4d, e). This might be caused by the reason that some dynamic chromatin changes were too subtle to identify. Alternatively, there may be a lag in the timing in the changes of the chromatin accessibility and the changes in gene expression (Potter *et al*., 2018).

Numerous studies have shown that RING E3 ubiquitin ligases play crucial roles in plant hormone signalling pathways, defence responses, and development processes by mediating protein stability (Park *et al*., 2012). DHS, a RING-type E3 ligase, degrades ROC4 to fine-tune wax biosynthesis (Wang *et al*., 2018). The RING-type E3 ligase MYB30-INTERACTING E3 LIGASE 1 (MIEL1) promotes MYB96 turnover to regulate ABA sensitivity in Arabidopsis (Lee and Seo, 2016). In apples, MdMIEL1 has been found to interact with and degrade MdBBX7 to regulate drought and cold tolerance (An *et al*., 2021; Chen *et al*., 2022). A point-to-point Y2H assay showed that the region containing the RING domain of MdRAD5B was responsible for its interaction with MdLHP1 (Figure 5a–c). Furthermore, our study showed that the degradation of MdLHP1 by MdRAD5B was through a 20S proteasome instead of a ubiquitin-dependent 26S proteasome (Figure 5i–k; Figure S8d-h). To our surprise, the interaction of MdPBA1 with MdRAD5B or MdLHP1 could not occur *in vitro* (Figure S8h), indicating that other subunits in the ubiquitin-independent 20S core proteasome might be directly involved in the degradation of MdLHP1 by MdRAD5B. Moreover, the increased ubiquitination of Myc-MdLHP1 by MG132 suggests that the degradation of MdLHP1 also occurs through the 26S ubiquitin-dependent pathway (Figure 5i). However, this 26S ubiquitin-dependent degradation was not due to MdRAD5B.

H3K27me3 regulates plant development and plant adaptation to environmental cues (Shen *et al*., 2021). H3K27me3 and its epigenetic silencing of the floral inhibitor *FLC* act as the central hub in the control of vernalization (Berry and Dean, 2015). Recent work showed that genes involved in response to water deprivation are targeted by H3K27me3 in *bli* mutants (Kleinmanns *et al*., 2017). Therefore, protein stability regulation of MdLHP1 by MdRAD5B implies a possible gene expression modulation of MdRAD5B through MdLHP1. Indeed, the H3K27me3 level was affected in *MdRAD5B* transgenic plants (Figure 6a,b). Specifically, the expression level of 66 DEGs was influenced by differentially H3K27me3 deposition in *MdRAD5B* RNAi transgenic plants under dehydration conditions, as compared with GL-3 plants (Figure 6e). In addition, genetic interaction analysis suggested that MdLHP1 acts downstream of MdRAD5B in drought response (Figure 7; Figure S21). These results suggest that MdRAD5B could regulate the expression of drought-responsive genes by regulating H3K27me3 deposition in response to drought stress, and this regulation might occur through MdLHP1. There were still many H3K27me3 hypo-methylated genes in *MdRAD5B* RNAi transgenic plants under control and dehydration conditions as compared with GL-3 plants (Figure 6e; Data S11 and S12), consistent with the presence of H3K27me3 hyper-methylated genes in *lhp1* (Veluchamy *et al*., 2016), indicating that there might be an unknown way for MdLHP1 to decrease the H3K27me3 disposition of some regions under control and dehydration conditions.

In summary, we demonstrated that MdRAD5B modulates the expression level of drought-responsive genes through its chromatin remodelling function or by regulating H3K27me3 in response to drought stress (Figure S24). Our work not only provides new evidence for the importance of chromatin remodelling factors in plant drought response but also benefits genetic engineering of fruit crops in the future.

Experimental procedures

Plant materials and growth conditions

Genes were cloned from the leaves of an apple (*Malus domestica* cv. Golden Delicious). GL-3 plants with high regeneration capacity isolated from 'Royal Gala' (*Malus domestica*) were used as the background of genetic transformation (Dai *et al*., 2013). Four-week-old tissue-cultured GL-3 plants cultivated on MS medium, which contained MS salts (4.43 g/L), sucrose (30 g/L), 6-BA (0.2 mg/L), IAA (0.2 mg/L), and agar (7.5 g/L) with a pH of 5.8–6.0 under long-day conditions (14 h light and 10 h dark) at 25 °C were used for stable transformation as previously described (Xie *et al*., 2018). Transgenic plants, as well as GL-3 plants, were rooted in MS medium, which contained MS salts (2.23 g/L), sucrose (20 g/L), IBA (0.5 mg/L), IAA (0.5 mg/L), and agar (7.5 g/L) with a pH of 5.8–6.0 under long-day conditions (14 h light and 10 h dark) at 25 °C for 1 month. Then, GL-3 and *MdRAD5B* transgenic plants were transferred into pots (12 × 13 cm) with long-day conditions (14 h light and 10 h dark) at 25 °C.

Apple calli (*Malus domestica* cv. Orin) were cultivated on MS agar medium, which contained MS salts (4.43 g/L), sucrose (30 g/L), 6-BA (0.4 mg/L), 2,4-D (1.5 mg/L), and agar (7.5 g/L) with a pH of 5.8–6.0 in darkness at 24 °C.

Stress treatment

Drought treatment was performed as described previously (Li *et al*., 2020a). Briefly, for short-term treatment, 2-month-old GL-3 and *MdRAD5B* transgenic plants grown in soil were divided into two groups, well-watered group ($n = 27$) and the drought group ($n = 27$). For the drought group, plants were withheld with water for 14 or 16 days, recovered by rewatering for 7 days, and then survival rate was calculated. For each treatment, 9 plants were used as one replicate.

For dehydration treatment, leaves were detached from 2-month-old GL-3 and *MdRAD5B* transgenic plants and dried in the air for 0 or 2 h. For each treatment, 4 leaves were used as one biological replicate, and 3 biological replicates were used.

PEG treatment was carried out as described previously (Chen *et al*., 2022). Equal weight 2-week-old wild type or transgenic calli grown on MS agar medium were transferred to MS agar medium containing 1.2 g/L MES [2-(N-morpholino) ethanesulphonic acid] or PEG8000 (Sigma, P2139, USA) to simulate low water potential (−0.7 MPa). Fresh weight was calculated after 14 days of PEG treatment. Experiments were performed for at least 2 times, and 5 plates were used for each treatment.

Generation of transgenic apple plants and apple calli

The coding region (CDS) of *MdRAD5B* was cloned into the pK2GW7 vector to generate an overexpressing plasmid. To knock down *MdRAD5B*, a 256-bp fragment of *MdRAD5B* was inserted into the pK7WIWG2D vector. The resulting plasmids were transformed into *Agrobacterium* strain *EHA105*, then to GL-3 plants by *Agrobacterium*-mediated transformation (Xie *et al*., 2018).

The transformation of apple calli was performed as described previously (An *et al*., 2021). The same plasmids that we generated for apple plants transformation were used to overexpress or knock down *MdRAD5B* in calli. To obtain *MdLHP1* anti calli, a 150-bp antisense fragment of *MdLHP1* was constructed into pH2GW7. In addition, the *MdLHP1*-pH2GW7 plasmid was also introduced into *MdRAD5B* RNAi calli to generate transgenic calli repressing both *MdRAD5B* and *MdLHP1*. The primers used are listed in Table S2.

ATPase activity assay

The ATPase activities of MdRAD5B and its mutant were detected as previously described (Guan *et al*., 2013). First, a fragment of MdRAD5B cDNA encoding the DEXDc region (amino acids 371–800 from the N-terminal region) was amplified and fused with a GST-tag by cloning into pET42a(+). The resulting pET42a(+)-MdRAD5B vector was used as the template to obtain a mutant version of MdRAD5B (Lysine at the 562 amino acids from the N-terminal region changed to Alanine) by site-directed mutagenesis, resulting in pET42a(+)-MdRAD5B-K562A plasmid. The two vectors were individually transformed into *Escherichia coli* Rosetta (DE3) pLysS cells, and the fusion proteins were induced with 0.8 nM IPTG at 28 °C and 120 r/min for 8 h. The induced proteins were then purified with glutathione S-transferase affinity columns (GE Healthcare) and used for the ATPase activity assay. Each reaction volume was 0.2 mL containing 50 mM KCl, 20 mM Tris–HCl, 5 mM $MgCl_2$, 2 mM phosphoenolpyruvate, 1 mM DTT, 1 mM ATP, 300 μM NADH, 50 nM MdRAD5B (or MdRAD5B-K562A), and 3 units/mL of pyruvate kinase and lactate dehydrogenase with or without 50 μg/mL *Malus domestica* total DNA. The reaction was performed at 28 °C, followed by the measurement of the absorbance at 338 nm. Three biological replicates were used.

Total histone protein blot

To detect histone modification levels, total proteins from GL-3 and transgenic plants with different treatments were extracted with 3× SDS loading buffer as described previously (Lu *et al*., 2011). Briefly, 0.1 g leaf samples were used for protein extraction, followed by boiling at 99 °C for 10 min. After centrifugation for 5 min at 13 523 *g*, the supernatants were then used for western blot. The H3K27me3 was detected by a α-H3K27me3 antibody. α-H3 (Pioneer in Proteomics, PTM-1002, China) immunoblot was used as a loading control. The experiments were repeated for at least three times.

Acknowledgements

We thank Dr. Zhihong Zhang from Shenyang Agricultural University for providing GL-3 plants. We thank Hight Performance Computing (HPC) platform of Northwest A&F University (NWAFU) for providing computing resources.

Conflict of interest

The authors declare that they have no competing interests.

Author contributions

Q.G. and Y.S. conceived and designed this study. Y.S. performed most of the experiments. J.H. analysed the RNA-seq, ATAC-seq and ChIP-seq data. J.G., Y.X., and Y.S. conducted the GL-3 and calli transformation. Z.M., Z.L., C.N., X.L., and B.C. took part in the plant and calli cultivation. Q.G., Y.S., J.H., and J.X. drafted the manuscript. F.M. helped with the discussion. M.T. improved the language of the manuscript. Q.G. supervized the study.

Funding

This work was supported by the National Nature Science Foundation of China (31872080), the Key S&T Special Projects of Shaanxi Province, China (2020zdzx03-01-02), and the Central Funds Guiding the Local Science and Technology Development of Shenzhen (2021SZVUP117), and Natural Science Foundation of Shaanxi Province, China (2022JQ-179).

Data availability statement

ATAC-seq, RNA-seq and ChIP-seq data were submitted to the Sequence Read Archive of the NCBI under the accession number PRJNA898595.

References

Abid, M., Tian, Z., Ata-Ul-Karim, S.T., Cui, Y., Liu, Y., Zahoor, R., Jiang, D. *et al*. (2016) Nitrogen nutrition improves the potential of wheat (*Triticum aestivum* L.) to alleviate the effects of drought stress during vegetative growth periods. *Front. Plant Sci.* **7**, 981.

An, J.P., Wang, X.F., Zhang, X.W., You, C.X. and Hao, Y.J. (2021) Apple B-box protein BBX37 regulates jasmonic acid mediated cold tolerance through the JAZ-BBX37-ICE1-CBF pathway and undergoes MIEL1-mediated ubiquitination and degradation. *New Phytol.* **229**, 2707–2729.

Asensi-Fabado, M.A., Amtmann, A. and Perrella, G. (2017) Plant responses to abiotic stress: the chromatin context of transcriptional regulation. *Biochim. Biophys. Acta Gene Regul. Mech.* **1860**, 106–122.

Bao, Y., Song, W.M., Pan, J., Jiang, C.M., Srivastava, R., Li, B., Zhu, L.Y. *et al*. (2016) Overexpression of the NDR1/HIN1-Like gene NHL6 modifies seed germination in response to abscisic acid and abiotic stresses in Arabidopsis. *PLoS One*, **11**, e0148572.

Bard, J.A.M., Goodall, E.A., Greene, E.R., Jonsson, E., Dong, K.C. and Martin, A. (2018) Structure and function of the 26S proteasome. *Annu. Rev. Biochem.* **87**, 697–724.

Berry, S. and Dean, C. (2015) Environmental perception and epigenetic memory: mechanistic insight through FLC. *Plant J.* **83**, 133–148.

Bo, K., Wang, H., Pan, Y., Behera, T.K., Pandey, S., Wen, C., Wang, Y. *et al*. (2016) SHORT HYPOCOTYL1 encodes a SMARCA3-like chromatin remodeling factor regulating elongation. *Plant Physiol.* **172**, 1273–1292.

Book, A.J., Gladman, N.P., Lee, S.S., Scalf, M., Smith, L.M. and Vierstra, R.D. (2010) Affinity purification of the Arabidopsis 26S proteasome reveals a diverse array of plant proteolytic complexes. *J. Biol. Chem.* **285**, 25554–25569.

Chen, I.P., Mannuss, A., Orel, N., Heitzeberg, F. and Puchta, H. (2008) A homolog of ScRAD5 is involved in DNA repair and homologous recombination in Arabidopsis. *Plant Physiol.* **146**, 1786–1796.

Chen, M., Ni, L., Chen, J., Sun, M., Qin, C., Zhang, G., Zhang, A. *et al*. (2021) Rice calcium/calmodulin-dependent protein kinase directly phosphorylates a mitogen-activated protein kinase kinase to regulate abscisic acid responses. *Plant Cell*, **33**, 1790–1812.

Chen, P., Yan, M., Li, L., He, J., Zhou, S., Li, Z., Niu, C. *et al*. (2020) The apple DNA-binding one zinc-finger protein MdDof54 promotes drought resistance. *Hortic Res.* **7**, 195.

Chen, P., Zhi, F., Li, X., Shen, W., Yan, M., He, J., Bao, C. *et al*. (2022) Zinc-finger protein MdBBX7/MdCOL9, a target of MdMIEL1 E3 ligase, confers drought tolerance in apple. *Plant Physiol.* **188**, 540–559.

Clapier, C.R. and Cairns, B.R. (2009) The biology of chromatin remodeling complexes. *Annu. Rev. Biochem.* **78**, 273–304.

Clapier, C.R., Iwasa, J., Cairns, B.R. and Peterson, C.L. (2017) Mechanisms of action and regulation of ATP-dependent chromatin-remodelling complexes. *Nat. Rev. Mol. Cell Biol.* **18**, 407–422.

Dai, H., Li, W., Han, G., Yi, Y. and Zhang, Z.J.S.H. (2013) Development of a seedling clone with high regeneration capacity and susceptibility to *Agrobacterium* in apple. *Sci. Hortic.* **164**, 202–208.

Derkacheva, M., Steinbach, Y., Wildhaber, T., Mozgova, I., Mahrez, W., Nanni, P., Bischof, S. *et al*. (2013) Arabidopsis MSI1 connects LHP1 to PRC2 complexes. *EMBO J.* **32**, 2073–2085.

Deshaies, R.J. and Joazeiro, C.A. (2009) RING domain E3 ubiquitin ligases. *Annu. Rev. Biochem.* **78**, 399–434.

Dong, T., Xu, Z.Y., Park, Y., Kim, D.H., Lee, Y. and Hwang, I. (2014) Abscisic acid uridine diphosphate glucosyltransferases play a crucial role in abscisic acid homeostasis in Arabidopsis. *Plant Physiol.* **165**, 277–289.

Duan, C.G., Zhu, J.K. and Cao, X.F. (2018) Retrospective and perspective of plant epigenetics in China. *J. Genet. Genomics*, **45**, 621–638.

Fujita, Y., Nakashima, K., Yoshida, T., Katagiri, T., Kidokoro, S., Kanamori, N., Umezawa, T. *et al*. (2009) Three SnRK2 protein kinases are the main positive regulators of abscisic acid signaling in response to water stress in Arabidopsis. *Plant Cell Physiol.* **50**, 2123–2132.

Gan, E.S., Huang, J. and Ito, T. (2013) Functional roles of histone modification, chromatin remodeling and microRNAs in Arabidopsis flower development. *Int. Rev. Cell Mol. Biol.* **305**, 115–161.

Gan, E.S., Xu, Y. and Ito, T. (2015) Dynamics of H3K27me3 methylation and demethylation in plant development. *Plant Signal. Behav.* **10**, e1027851.

Geng, D., Chen, P., Shen, X., Zhang, Y., Li, X., Jiang, L., Xie, Y. *et al*. (2018) MdMYB88 and MdMYB124 enhance drought tolerance by modulating root vessels and cell walls in apple. *Plant Physiol.* **178**, 1296–1309.

Guan, Q., Wu, J., Zhang, Y., Jiang, C., Liu, R., Chai, C. and Zhu, J. (2013) A DEAD box RNA helicase is critical for pre-mRNA splicing, cold-responsive gene regulation, and cold tolerance in Arabidopsis. *Plant Cell*, **25**, 342–356.

Guillaumot, D., Guillon, S., Déplanque, T., Vanhee, C., Gumy, C., Masquelier, D., Morsomme, P. *et al*. (2009) The Arabidopsis TSPO-related protein is a stress and abscisic acid-regulated, endoplasmic reticulum-Golgi-localized membrane protein. *Plant J.* **60**, 242–256.

Gupta, A., Rico-Medina, A. and Caño-Delgado, A.I. (2020) The physiology of plant responses to drought. *Science*, **368**, 266–269.

Hou, N., Li, C., He, J., Liu, Y., Yu, S., Malnoy, M., Mobeen Tahir, M. *et al*. (2022) MdMTA-mediated m(6) A modification enhances drought tolerance by promoting mRNA stability and translation efficiency of genes involved in lignin deposition and oxidative stress. *New Phytol.* **234**, 1294–1314.

Hu, Y., Zhu, N., Wang, X., Yi, Q., Zhu, D., Lai, Y. and Zhao, Y. (2013) Analysis of rice Snf2 family proteins and their potential roles in epigenetic regulation. *Plant Physiol. Biochem*. **70**, 33–42.

Iyer, L.M., Babu, M.M. and Aravind, L. (2006) The HIRAN domain and recruitment of chromatin remodeling and repair activities to damaged DNA. *Cell Cycle*, **5**, 775–782.

Jégu, T., Veluchamy, A., Ramirez-Prado, J.S., Rizzi-Paillet, C., Perez, M., Lhomme, A., Latrasse, D. *et al*. (2017) The Arabidopsis SWI/SNF protein BAF60 mediates seedling growth control by modulating DNA accessibility. *Genome Biol.* **18**, 114.

Jia, X., Gong, X., Jia, X., Li, X., Wang, Y., Wang, P., Huo, L. *et al*. (2021) Overexpression of MdATG8i enhances drought tolerance by alleviating oxidative damage and promoting water uptake in transgenic apple. *Int. J. Mol. Sci.* **22**, 5517.

Jiang, L., Zhang, D., Liu, C., Shen, W., He, J., Yue, Q., Niu, C. *et al*. (2022) MdGH3.6 is targeted by MdMYB94 and plays a negative role in apple water-deficit stress tolerance. *Plant J.* **109**, 1271–1289.

Kang, H., Liu, Y., Fan, T., Ma, J., Wu, D., Heitz, T., Shen, W.H. *et al*. (2022) Arabidopsis CHROMATIN REMODELING 19 acts as a transcriptional repressor and contributes to plant pathogen resistance. *Plant Cell*, **34**, 1100–1116.

Kang, J.Y., Choi, H.I., Im, M.Y. and Kim, S.Y. (2002) Arabidopsis basic leucine zipper proteins that mediate stress-responsive abscisic acid signaling. *Plant Cell*, **14**, 343–357.

Khatri, N., Singh, S., Hakim, N. and Mudgil, Y. (2017) Comparative expression profiling of AtRAD5B and AtNDL1: hints towards a role in G protein mediated signaling. *Gene Expr. Patterns*, **25–26**, 167–174.

Kim, J.H. (2021) Multifaceted chromatin structure and transcription changes in plant stress response. *Int. J. Mol. Sci.* **22**, 2013.

Kim, S., Kang, J.Y., Cho, D.I., Park, J.H. and Kim, S.Y. (2004) ABF2, an ABRE-binding bZIP factor, is an essential component of glucose signaling and its overexpression affects multiple stress tolerance. *Plant J.* **40**, 75–87.

Kleinmanns, J.A., Schatlowski, N., Heckmann, D. and Schubert, D. (2017) BLISTER regulates polycomb-target genes, represses stress-regulated genes and promotes stress responses in *Arabidopsis thaliana*. *Front. Plant Sci.* **8**, 1530.

Knizewski, L., Ginalski, K. and Jerzmanowski, A. (2008) Snf2 proteins in plants: gene silencing and beyond. *Trends Plant Sci.* **13**, 557–565.

Kotake, T., Takada, S., Nakahigashi, K., Ohto, M. and Goto, K. (2003) Arabidopsis TERMINAL FLOWER 2 gene encodes a heterochromatin protein 1 homolog and represses both FLOWERING LOCUS T to regulate flowering time and several floral homeotic genes. *Plant Cell Physiol.* **44**, 555–564.

Kumar, R., Li, D.Q., Müller, S. and Knapp, S. (2016) Epigenomic regulation of oncogenesis by chromatin remodeling. *Oncogene*, **35**, 4423–4436.

Lee, H.G. and Seo, P.J. (2016) The Arabidopsis MIEL1 E3 ligase negatively regulates ABA signalling by promoting protein turnover of MYB96. *Nat. Commun.* **7**, 12525.

Leitão, S.T., Santos, C., Araújo, S.S., Rubiales, D. and Vaz Patto, M.C. (2021) Shared and tailored common bean transcriptomic responses to combined fusarium wilt and water deficit. *Hortic. Res.* **8**, 149.

Li, X., Chen, P., Xie, Y., Yan, Y., Wang, L., Dang, H., Zhang, J. *et al*. (2020a) Apple SERRATE negatively mediates drought resistance by regulating MdMYB88 and MdMYB124 and microRNA biogenesis. *Hortic. Res.* **7**, 98.

Li, X., Zhou, S., Liu, Z., Lu, L., Dang, H., Li, H., Chu, B. *et al*. (2022a) Fine-tuning of SUMOylation modulates drought tolerance of apple. *Plant Biotechnol. J.* **20**, 903–919.

Li, Y., Sun, D., Ma, Z., Yamaguchi, K., Wang, L., Zhong, S., Yan, X. *et al*. (2020b) Degradation of SERRATE via ubiquitin-independent 20S proteasome to survey RNA metabolism. *Nat. Plants*, **6**, 970–982.

Li, Z., Wang, L., He, J., Li, X., Hou, N., Guo, J., Niu, C. *et al*. (2022b) A chromosome-scale reference genome provides insights into the genetic origin and grafting-mediated stress tolerance of *Malus prunifolia*. *Plant Biotechnol. J.* **20**, 1015–1017.

Liao, X., Guo, X., Wang, Q., Wang, Y., Zhao, D., Yao, L., Wang, S. *et al*. (2017) Overexpression of MsDREB6.2 results in cytokinin-deficient developmental phenotypes and enhances drought tolerance in transgenic apple plants. *Plant J.* **89**, 510–526.

Libault, M., Tessadori, F., Germann, S., Snijder, B., Fransz, P. and Gaudin, V. (2005) The Arabidopsis LHP1 protein is a component of euchromatin. *Planta*, **222**, 910–925.

Liu, C., Lu, F., Cui, X. and Cao, X. (2010) Histone methylation in higher plants. *Annu. Rev. Plant Biol.* **61**, 395–420.

Liu, H., Guo, S., Lu, M., Zhang, Y., Li, J., Wang, W., Wang, P. *et al*. (2019) Biosynthesis of DHGA(12) and its roles in Arabidopsis seedling establishment. *Nat. Commun.* **10**, 1768.

Liu, X., Zhao, C., Gao, Y., Xu, Y., Wang, S., Li, C., Xie, Y. *et al*. (2021) A multifaceted module of BRI1 ETHYLMETHANE SULFONATE SUPRESSOR1 (BES1)-MYB88 in growth and stress tolerance of apple. *Plant Physiol.* **185**, 1903–1923.

Lu, F., Cui, X., Zhang, S., Jenuwein, T. and Cao, X. (2011) Arabidopsis REF6 is a histone H3 lysine 27 demethylase. *Nat. Genet.* **43**, 715–719.

Ma, Y., Cao, J., Chen, Q., He, J., Liu, Z., Wang, J., Li, X. *et al*. (2019) The kinase CIPK11 functions as a negative regulator in drought stress response in Arabidopsis. *Int. J. Mol. Sci.* **20**, 2422.

Medina, J., Rodríguez-Franco, M., Peñalosa, A., Carrascosa, M.J., Neuhaus, G. and Salinas, J. (2005) Arabidopsis mutants deregulated in RCI2A expression reveal new signaling pathways in abiotic stress responses. *Plant J.* **42**, 586–597.

Niu, C., Li, H., Jiang, L., Yan, M., Li, C., Geng, D., Xie, Y. *et al*. (2019) Genome-wide identification of drought-responsive microRNAs in two sets of Malus from interspecific hybrid progenies. *Hortic. Res.* **6**, 75.

Niu, Y., Wang, Y., Li, P., Zhang, F., Liu, H. and Zheng, G. (2013) Drought stress induces oxidative stress and the antioxidant defense system in ascorbate-deficient vtc1 mutants of *Arabidopsis thaliana*. *Acta Physiol. Plant.* **35**, 1189–1200.

Park, C.H., Chen, S., Shirsekar, G., Zhou, B., Khang, C.H., Songkumarn, P., Afzal, A.J. *et al*. (2012) The Magnaporthe oryzae effector AvrPiz-t targets the RING E3 ubiquitin ligase APIP6 to suppress pathogen-associated molecular pattern-triggered immunity in rice. *Plant Cell*, **24**, 4748–4762.

Park, S.C., Kim, Y.H., Jeong, J.C., Kim, C.Y., Lee, H.S., Bang, J.W. and Kwak, S.S. (2011) Sweetpotato late embryogenesis abundant 14 (IbLEA14) gene influences lignification and increases osmotic- and salt stress-tolerance of transgenic calli. *Planta*, **233**, 621–634.

Potter, K.C., Wang, J., Schaller, G.E. and Kieber, J.J. (2018) Cytokinin modulates context-dependent chromatin accessibility through the type-B response regulators. *Nat. Plants*, **4**, 1102–1111.

Ramegowda, V., Gill, U.S., Sivalingam, P.N., Gupta, A., Gupta, C., Govind, G., Nataraja, K.N. *et al*. (2017) GBF3 transcription factor imparts drought tolerance in *Arabidopsis thaliana*. *Sci. Rep.* **7**, 9148.

Ramirez-Prado, J.S., Latrasse, D., Rodriguez-Granados, N.Y., Huang, Y., Manza-Mianza, D., Brik-Chaouche, R., Jaouannet, M. *et al*. (2019) The Polycomb protein LHP1 regulates *Arabidopsis thaliana* stress responses through the repression of the MYC2-dependent branch of immunity. *Plant J.* **100**, 1118–1131.

Rao, M.J., Xu, Y., Tang, X., Huang, Y., Liu, J., Deng, X. and Xu, Q. (2020) CsCYT75B1, a citrus CYTOCHROME P450 gene, is involved in accumulation of antioxidant flavonoids and induces drought tolerance in transgenic Arabidopsis. *Antioxidants (Basel)*, **9**, 161.

Ré, D.A., Capella, M., Bonaventure, G. and Chan, R.L. (2014) Arabidopsis AtHB7 and AtHB12 evolved divergently to fine tune processes associated with growth and responses to water stress. *BMC Plant Biol.* **14**, 150.

Rinaldi, M.A., Liu, J., Enders, T.A., Bartel, B. and Strader, L.C. (2012) A gain-of-function mutation in IAA16 confers reduced responses to auxin and abscisic acid and impedes plant growth and fertility. *Plant Mol. Biol.* **79**, 359–373.

Ryan, D.P. and Owen-Hughes, T. (2011) Snf2-family proteins: chromatin remodellers for any occasion. *Curr. Opin. Chem. Biol.* **15**, 649–656.

Saneoka, H., Moghaieb, R., Premachandra, G.S. and Fujita, K. (2004) Nitrogen nutrition and water stress effects on cell membrane stability and leaf water relations in *Agrostis palustris* Huds. *Environ. Exp. Bot.* **52**, 131–138.

Sarnowska, E., Gratkowska, D.M., Sacharowski, S.P., Cwiek, P., Tohge, T., Fernie, A.R., Siedlecki, J.A. *et al*. (2016) The role of SWI/SNF chromatin remodeling complexes in hormone crosstalk. *Trends Plant Sci.* **21**, 594–608.

Sato, H., Suzuki, T., Takahashi, F., Shinozaki, K. and Yamaguchi-Shinozaki, K. (2019) NF-YB2 and NF-YB3 have functionally diverged and differentially induce drought and heat stress-specific genes. *Plant Physiol.* **180**, 1677–1690.

Shen, Q., Lin, Y., Li, Y. and Wang, G. (2021) Dynamics of H3K27me3 modification on plant adaptation to environmental cues. *Plants (Basel, Switzerland)*, **10**, 1165.

Shen, X., He, J., Ping, Y., Guo, J., Hou, N., Cao, F., Li, X. *et al*. (2022) The positive feedback regulatory loop of miR160-Auxin Response Factor 17-HYPONASTIC LEAVES 1 mediates drought tolerance in apple trees. *Plant Physiol*. **188**, 1686–1708.

Sullivan, A.M., Bubb, K.L., Sandstrom, R., Stamatoyannopoulos, J.A. and Queitsch, C. (2015) DNase I hypersensitivity mapping, genomic footprinting, and transcription factor networks in plants. *Curr. Plant Biol*. **3–4**, 40–47.

Sung, P., Higgins, D., Prakash, L. and Prakash, S. (1988) Mutation of lysine-48 to arginine in the yeast RAD3 protein abolishes its ATPase and DNA helicase activities but not the ability to bind ATP. *EMBO J*. **7**, 3263–3269.

Talbi, S., Romero-Puertas, M.C., Hernández, A., Terrón, L., Ferchichi, A. and Sandalio, L.M. (2015) Drought tolerance in a Saharian plant *Oudneya africana*: role of antioxidant defences. *Environ. Exp. Bot*. **111**, 114–126.

Tang, X., Fei, X., Sun, Y., Shao, H., Zhu, J., He, X., Wang, X. *et al*. (2022) Abscisic acid-polyacrylamide (ABA-PAM) treatment enhances forage grass growth and soil microbial diversity under drought stress. *Front. Plant Sci*. **13**, 973665.

Turck, F., Roudier, F., Farrona, S., Martin-Magniette, M.L., Guillaume, E., Buisine, N., Gagnot, S. *et al*. (2007) Arabidopsis TFL2/LHP1 specifically associates with genes marked by trimethylation of histone H3 lysine 27. *PLoS Genet*. **3**, e86.

Veluchamy, A., Jegu, T., Ariel, F., Latrasse, D., Mariappan, K.G., Kim, S.K., Crespi, M. *et al*. (2016) LHP1 regulates H3K27me3 spreading and shapes the three-dimensional conformation of the Arabidopsis genome. *PLoS One*, **11**, e0158936.

Vonapartis, E., Mohamed, D., Li, J., Pan, W., Wu, J. and Gazzarrini, S. (2022) CBF4/DREB1D represses XERICO to attenuate ABA, osmotic and drought stress responses in Arabidopsis. *Plant J*. **110**, 961–977.

Wang, H., Blakeslee, J.J., Jones, M.L., Chapin, L.J. and Dami, I.E. (2020) Exogenous abscisic acid enhances physiological, metabolic, and transcriptional cold acclimation responses in greenhouse-grown grapevines. *Plant Sci*. **293**, 110437.

Wang, Z., Tian, X., Zhao, Q., Liu, Z., Li, X., Ren, Y., Tang, J. *et al*. (2018) The E3 ligase DROUGHT HYPERSENSITIVE negatively regulates cuticular wax biosynthesis by promoting the degradation of transcription factor ROC4 in rice. *Plant Cell*, **30**, 228–244.

Xie, Y., Bao, C., Chen, P., Cao, F., Liu, X., Geng, D., Li, Z. *et al*. (2021) Abscisic acid homeostasis is mediated by feedback regulation of MdMYB88 and MdMYB124. *J. Exp. Bot*. **72**, 592–607.

Xie, Y., Chen, P., Yan, Y., Bao, C., Li, X., Wang, L., Shen, X. *et al*. (2018) An atypical R2R3 MYB transcription factor increases cold hardiness by CBF-dependent and CBF-independent pathways in apple. *New Phytol*. **218**, 201–218.

Xu, J., Zhou, S., Gong, X., Song, Y., van Nocker, S., Ma, F. and Guan, Q. (2018) Single-base methylome analysis reveals dynamic epigenomic differences associated with water deficit in apple. *Plant Biotechnol. J*. **16**, 672–687.

Yuan, L.Y., Liu, X.C., Luo, M., Yang, S.G. and Wu, K.Q. (2013) Involvement of histone modifications in plant abiotic stress responses. *J. Integr. Plant Biol*. **55**, 892–901.

Zhang, X., Germann, S., Blus, B.J., Khorasanizadeh, S., Gaudin, V. and Jacobsen, S.E. (2007) The Arabidopsis LHP1 protein colocalizes with histone H3 Lys27 trimethylation. *Nat. Struct. Mol. Biol*. **14**, 869–871.

Zhu, J.K. (2016) Abiotic stress signaling and responses in plants. *Cell*, **167**, 313–324.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1 Comparison of amino acid sequences of MdRAD5B with AtRAD5B.
Figure S2 Generation of *MdRAD5B* transgenic plants.
Figure S3 Phenotype of *MdRAD5B* transgenic plants under drought treatment.
Figure S4 The peaks of ATAC-seq and their genomic distributions in GL-3 and *MdRAD5B* RNAi plants under control and dehydration conditions.
Figure S5 Profiles and heatmaps of the peaks identified in GL-3 and *MdRAD5B* RNAi plants under control and dehydration conditions.
Figure S6 Altered chromatin accessibility in GL-3 plants in response to dehydration.
Figure S7 Integrative Genomics Viewer (IGV) tracks showing examples of significantly altered genes at chromatin accessibility and transcription level in *MdRAD5B* RNAi plants under dehydration conditions, as compared with GL-3 plants.
Figure S8 MdPBA1 interacts with MdLHP1 and MdRAD5B *in vivo*.
Figure S9 Subcellular localization of MdLHP1.
Figure S10 MdLHP1 interacts with MdMSI1 *in vivo*.
Figure S11 MdLHP1 can regulate H3K27me3 deposition in *Malus domestica*.
Figure S12 The peaks of ChIP-seq and their genomic distributions in GL-3 and *MdRAD5B* RNAi plants under control and dehydration conditions.
Figure S13 Profiles and heatmaps of the H3K27me3 histone modification in GL-3 and *MdRAD5B* RNAi plants under control and dehydration conditions.
Figure S14 Altered H3K27me3 histone modification in response to dehydration in GL-3 plants.
Figure S15 Integrative Genomics Viewer (IGV) tracks showing examples of significantly altered genes at the H3K27me3 abundance and transcription level in *MdRAD5B* RNAi plants under control and dehydration conditions, as compared with GL-3 plants.
Figure S16 The H3K27me3 abundance and transcription level of drought-responsive genes in *MdRAD5B* RNAi plants under control and dehydration conditions.
Figure S17 The H3K27me3 abundance of drought-responsive genes in *MdRAD5B* OE plants under control and dehydration conditions.
Figure S18 The transcription level of drought-responsive genes in *MdRAD5B* OE plants under control and dehydration conditions.
Figure S19 *MdRAD5B* expression in WT and transgenic *MdRAD5B* RNAi lines.
Figure S20 Generation of double transgenic calli of *MdLHP1* and *MdRAD5B*.
Figure S21 Genetic analysis of *MdRAD5B* and *MdLHP1*.
Figure S22 Generation of transgenic calli expressing different deletions of *MdRAD5B*.
Figure S23 Effects of HIRAN, DEXDc, and RING domain in MdRAD5B on simulated-drought tolerance.
Figure S24 A proposed model for *MdRAD5B* in response to drought stress.
Appendix S1 Supporting experimental procedures.
Data S1 RNA-seq information.
Data S2 Differentially expressed genes (DEGs) in *MdRAD5B* RNAi plants under control conditions, as compared with GL-3 plants.
Data S3 Differentially expressed genes (DEGs) in *MdRAD5B* RNAi plants under dehydration (DH) conditions, as compared with GL-3 plants.
Data S4 Differentially expressed genes (DEGs) in GL-3 plants under dehydration (DH) conditions, as compared with control conditions.
Data S5 Sequencing information of ATAC-seq.
Data S6 Differentially accessible regions (DARs) in GL-3 plants under dehydration (DH) conditions, as compared with control conditions.

Data S7 Differentially accessible regions (DARs) in *MdRAD5B* RNAi plants under control conditions, as compared with GL-3 plants.
Data S8 Differentially accessible regions (DARs) in *MdRAD5B* RNAi plants under dehydration (DH) conditions, as compared with GL-3 plants.
Data S9 ChIP-seq information.
Data S10 Differentially methylated (H3K27me3) regions in GL-3 plants under dehydration (DH) conditions, as compared with control conditions.
Data S11 Differentially methylated (H3K27me3) regions in *MdRAD5B* RNAi plants under control conditions, as compared with GL-3 plants.
Data S12 Differentially methylated (H3K27me3) regions in *MdRAD5B* RNAi plants under dehydration (DH) conditions, as compared with GL-3 plants.
Table S1 The potential interacting proteins of MdRAD5B.
Table S2 List of primers used in this study.